EE6108 Workshop Laboratory Manual

Course Title: Practical Computer Networking Workshop

Simulation Engine: Cisco Packet Tracer

Student Name: _____ **Student ID:** _____ **Group Number:** _____

Date: _____

Lab 1: Local Area Network Addressing & Connectivity Verification

1. Scenario & Objectives

In this introductory lab, you will configure a local area network (LAN) connecting two computers through a Layer 2 switch. You will practice network device placement, automatic cabling, static IPv4 configuration, and observe connectivity differences between same-subnet and cross-subnet setups.

- Deploy endpoint PCs and a Layer 2 switch using automatic connection cabling.
- Configure static IPv4 addresses and subnet masks (`/24`) through the device interface.
- Test and verify network reachability using the `ping` command line utility.

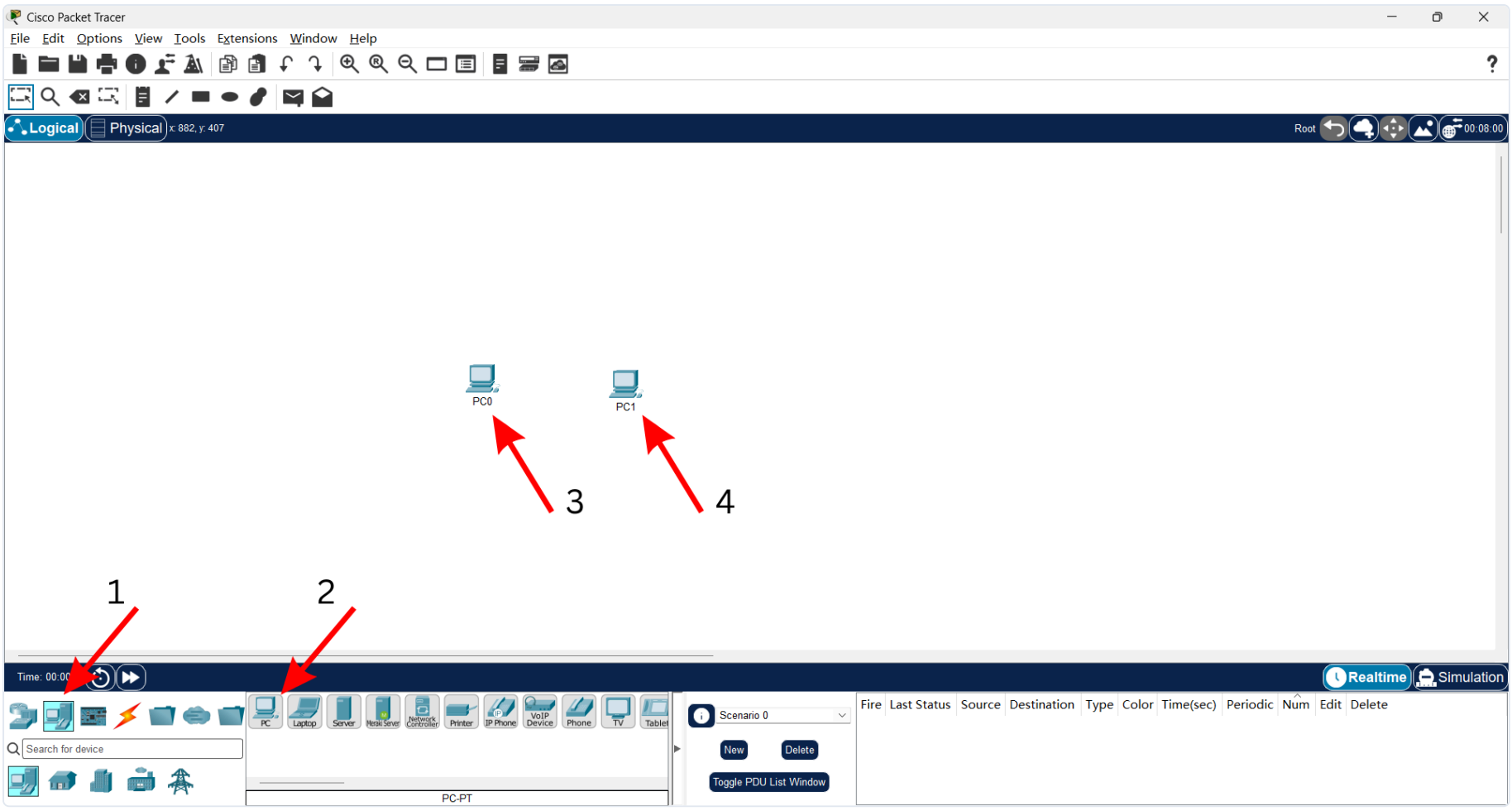
2. Addressing Table

Device	Interface	Initial Test IP	Target Corrected IP	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.1	192.168.1.1	255.255.255.0	None (Direct L2)
PC1	FastEthernet0	192.168.2.1	192.168.1.2	255.255.255.0	None (Direct L2)
Switch0	Fa0/1, Fa0/2	Unassigned	Unassigned	Layer 2 Transparent	N/A

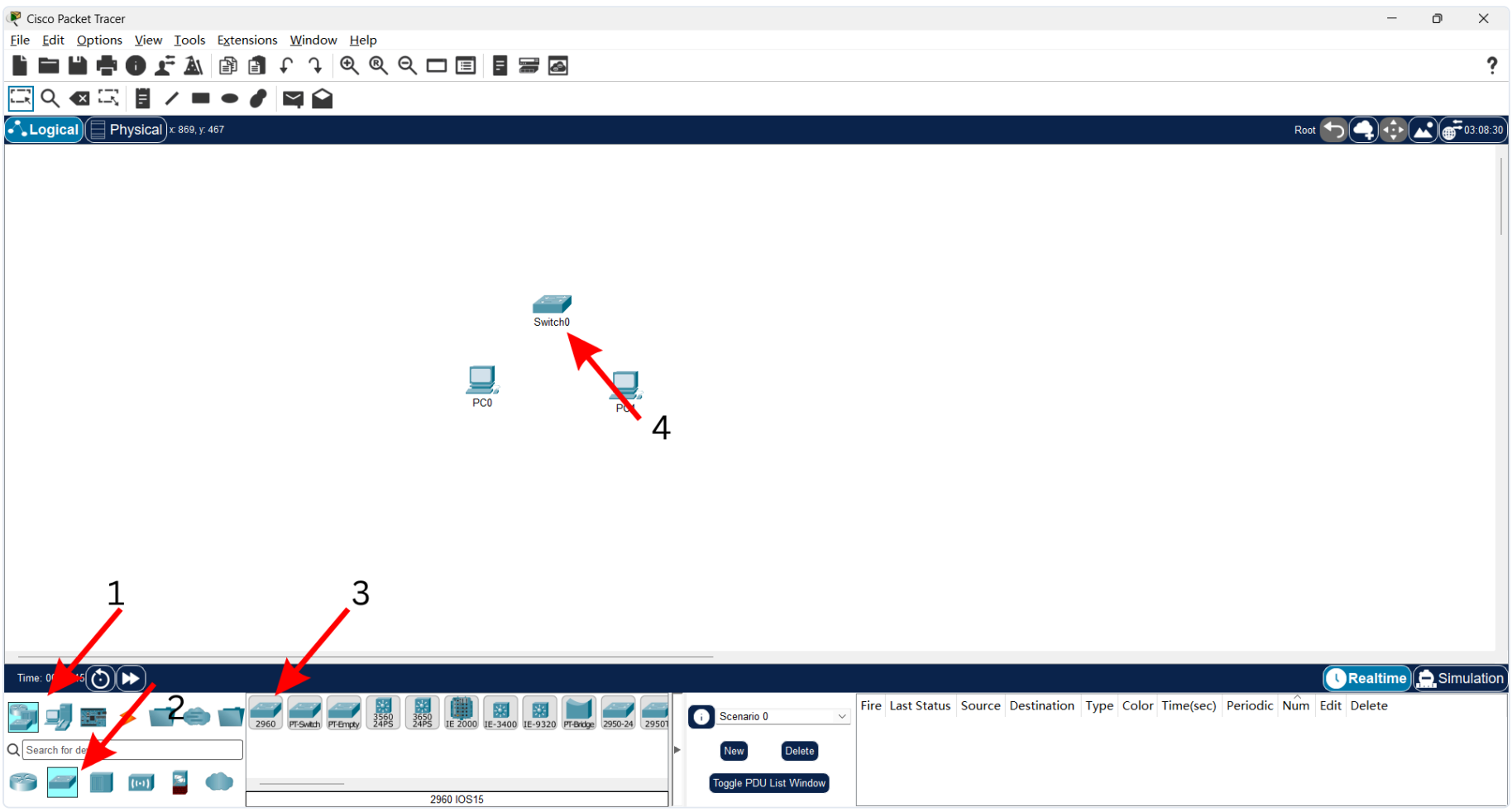
3. Step-by-Step Procedure

Step 1: Place Hardware Components and Connect Cable

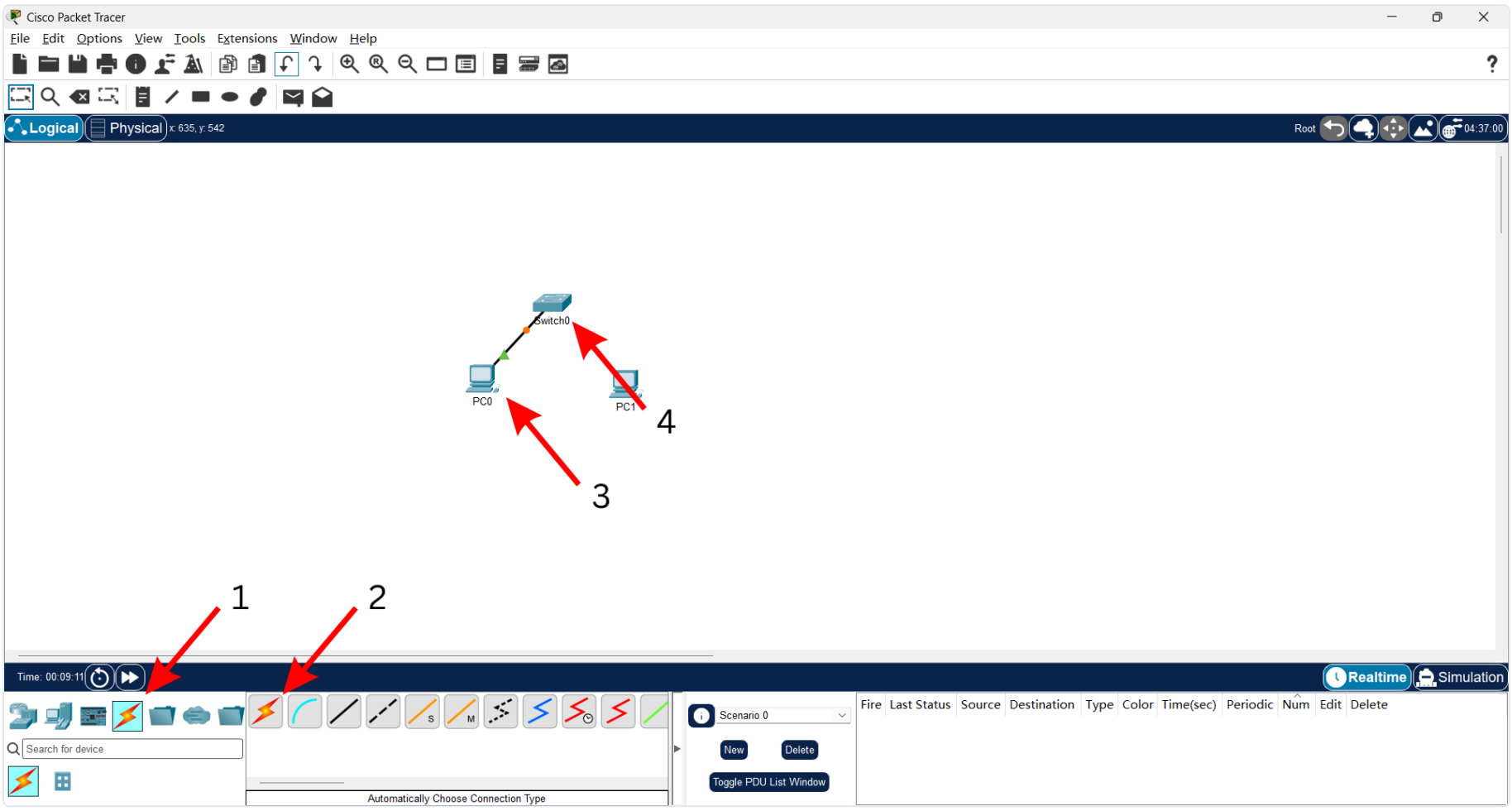
1. Navigate to the bottom-left menu, select **End Devices**, and drag two **PCs** (PC0, PC1) onto the workspace canvas.



2. Select **Network Devices** → **Switches**, and place one **Switch** (Switch0) above the two computers.

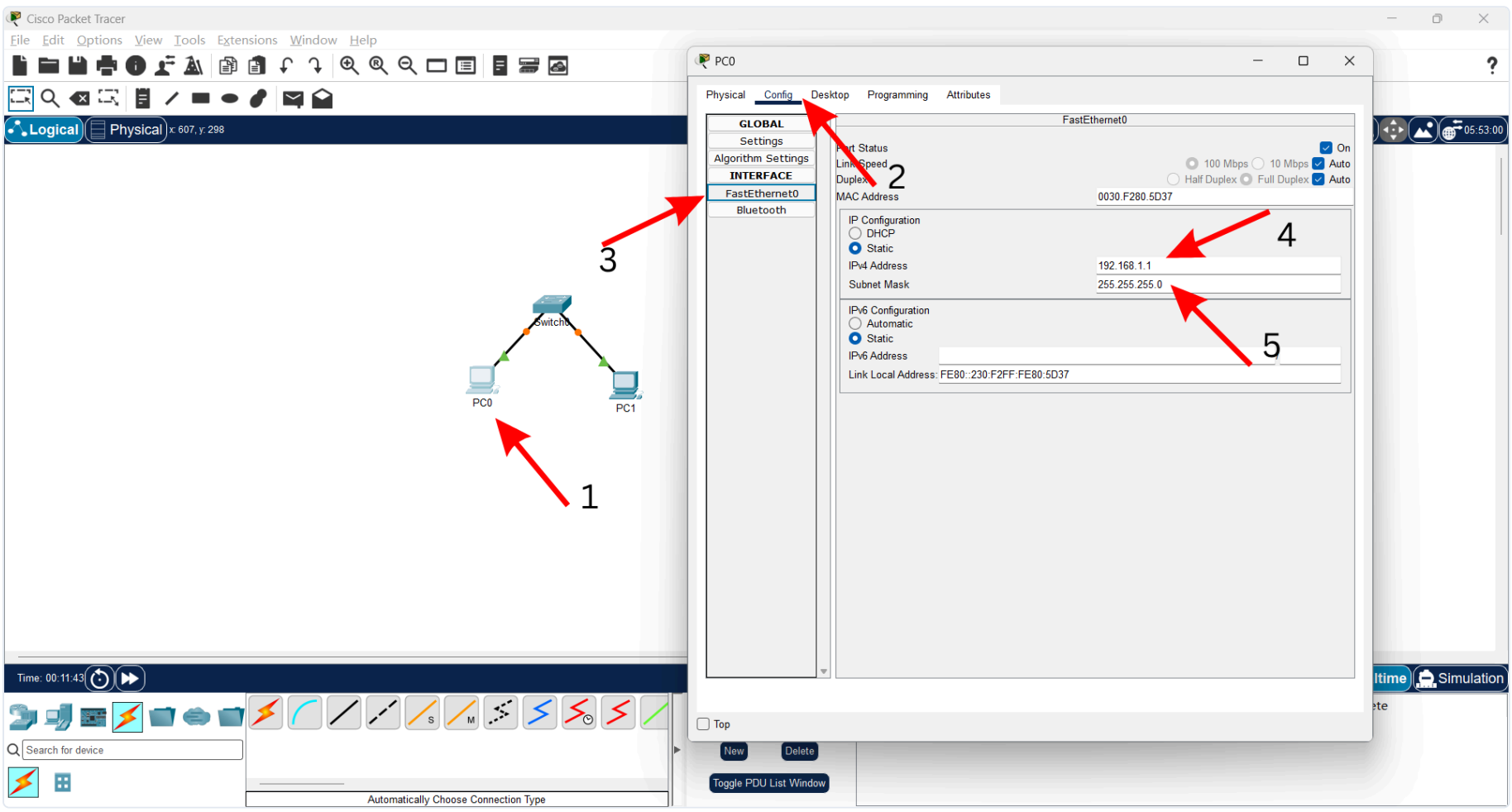


3. Click **Connections** → select **Automatically Choose Connection Type** (orange lightning bolt icon), then click PC0 and Switch0, and click PC1 and Switch0 to establish physical connections.

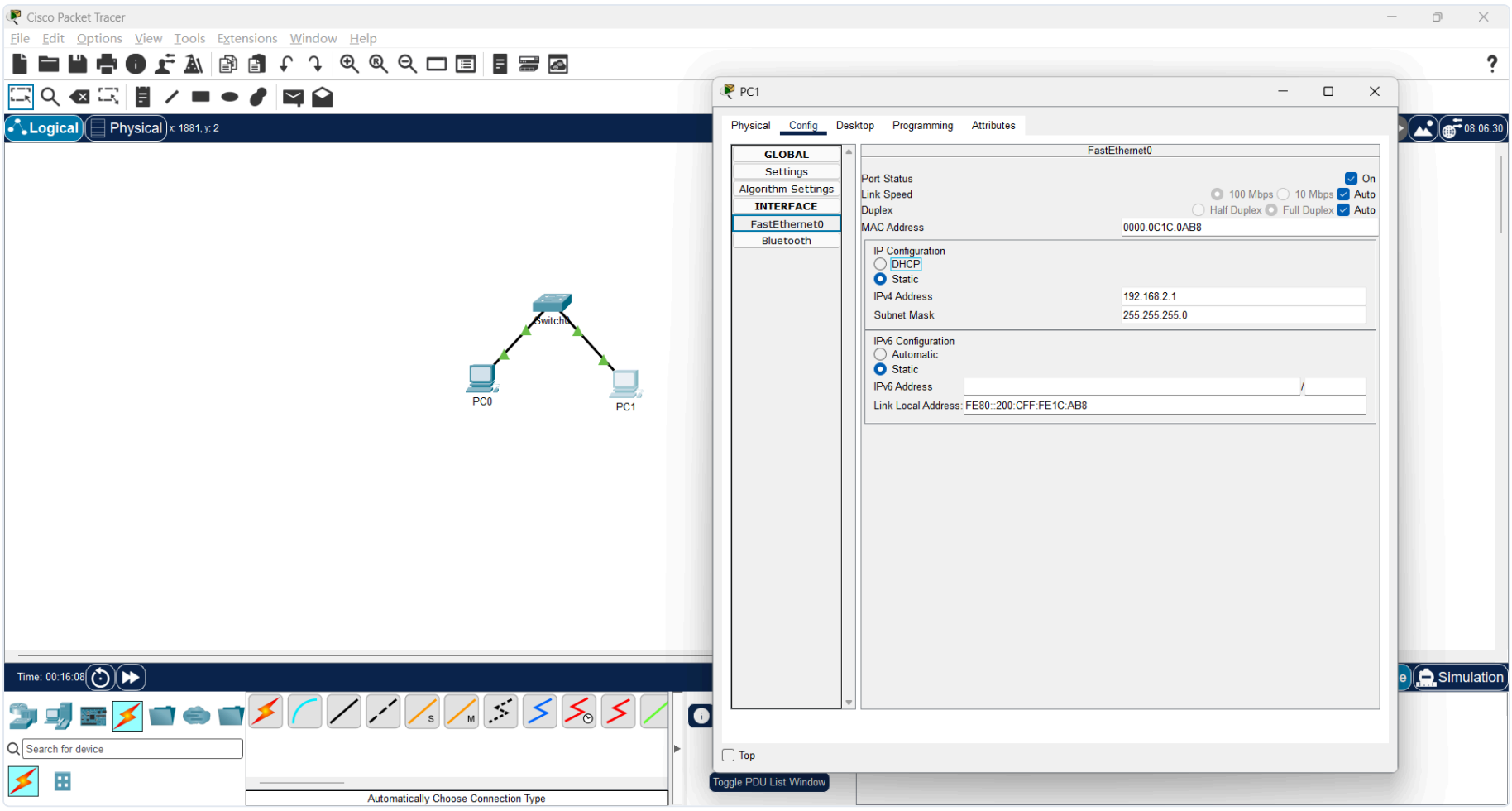


Step 2: Assign Initial Test IP Addresses

1. Click **PC0** → select the **Config** tab → click **FastEthernet0** (under INTERFACE):
    - IPv4 Address: `192.168.1.1`
    - Subnet Mask: `255.255.255.0`

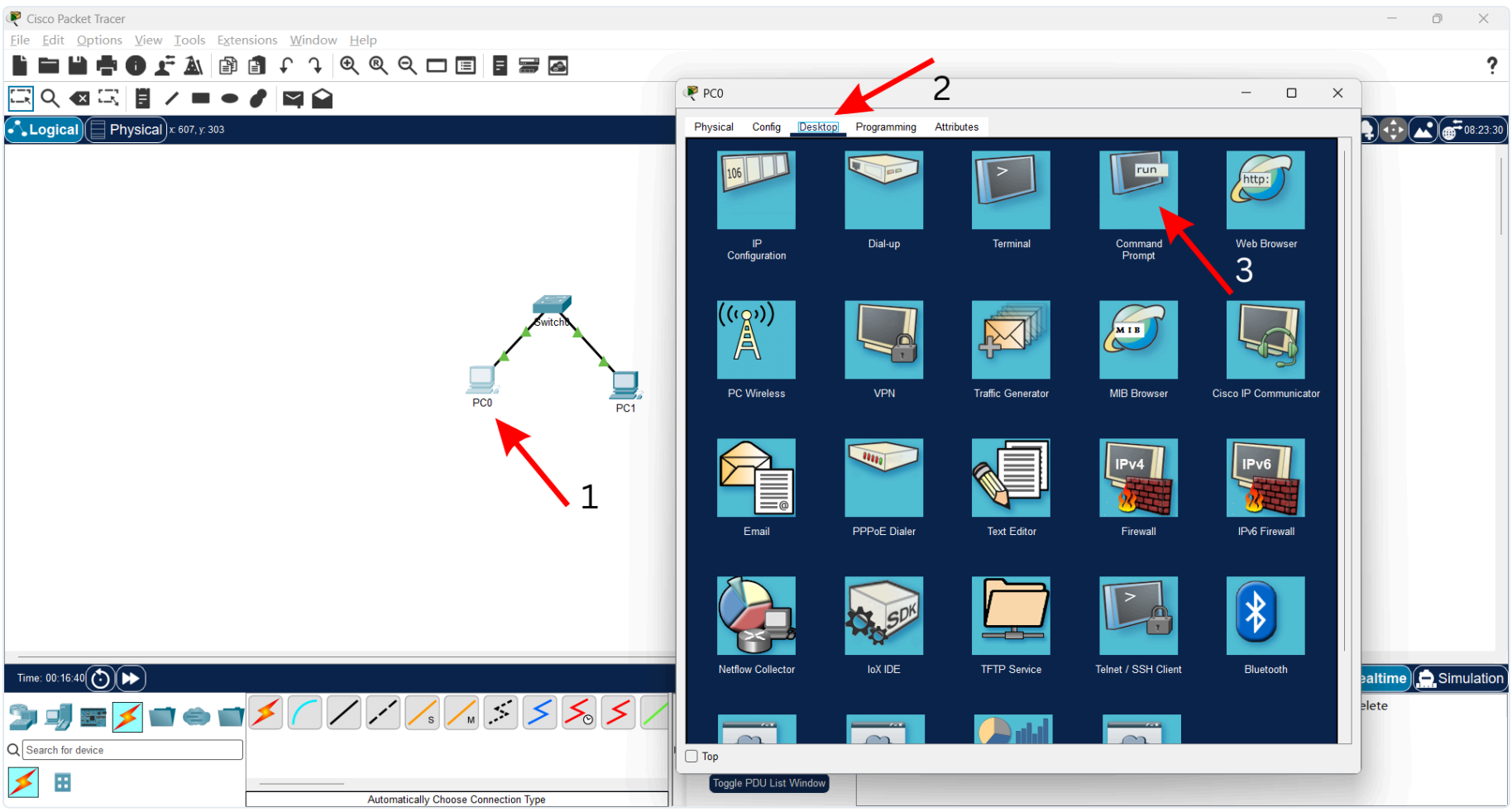


2. Click **PC1** → select the **Config** tab → click **FastEthernet0** (under INTERFACE):
    - IPv4 Address: `192.168.2.1`
    - Subnet Mask: `255.255.255.0`



Step 3: Test Initial Connectivity

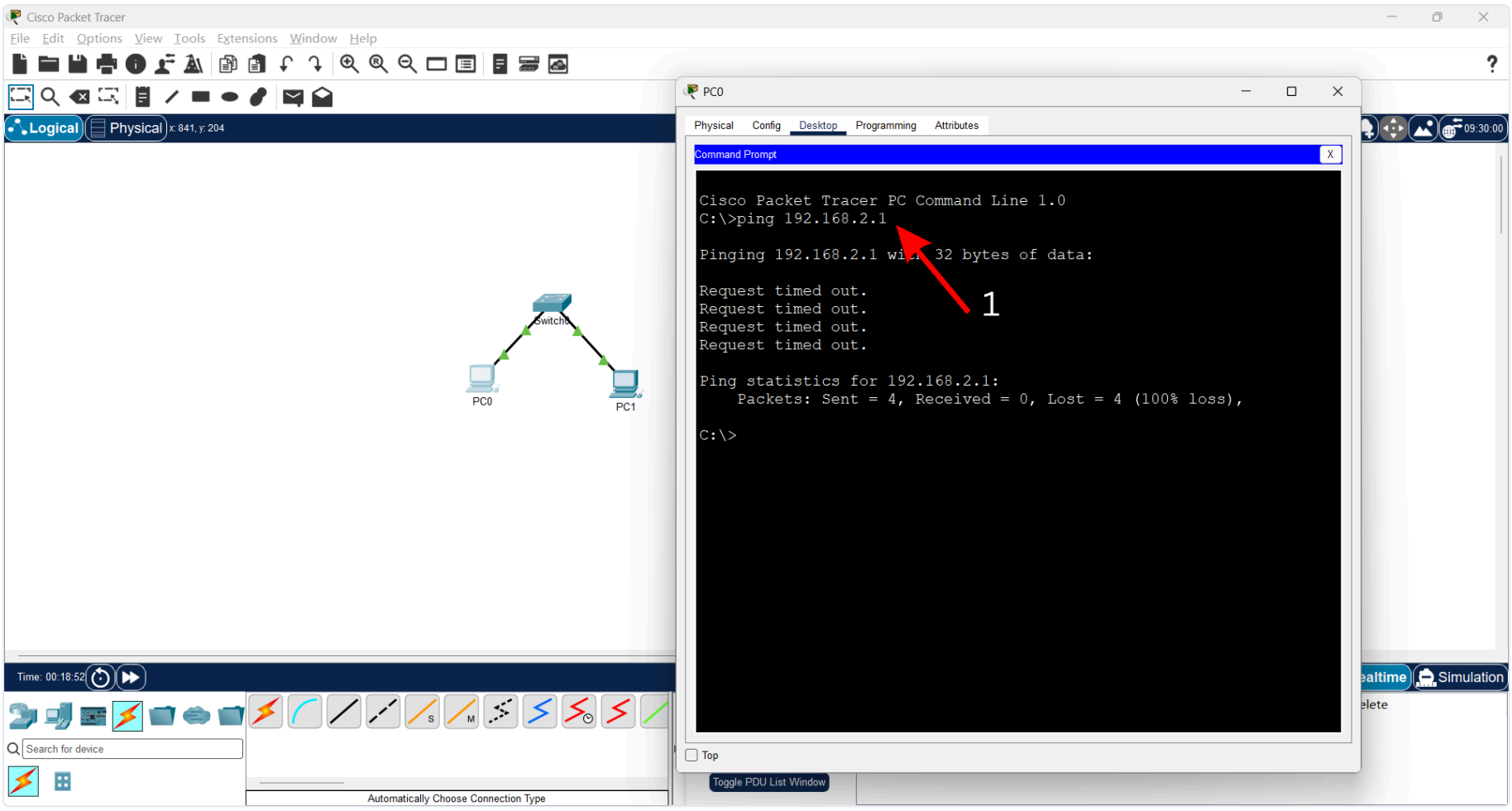
1. Click **PC0** → select the **Desktop** tab → open **Command Prompt**.



2. Execute the test command:

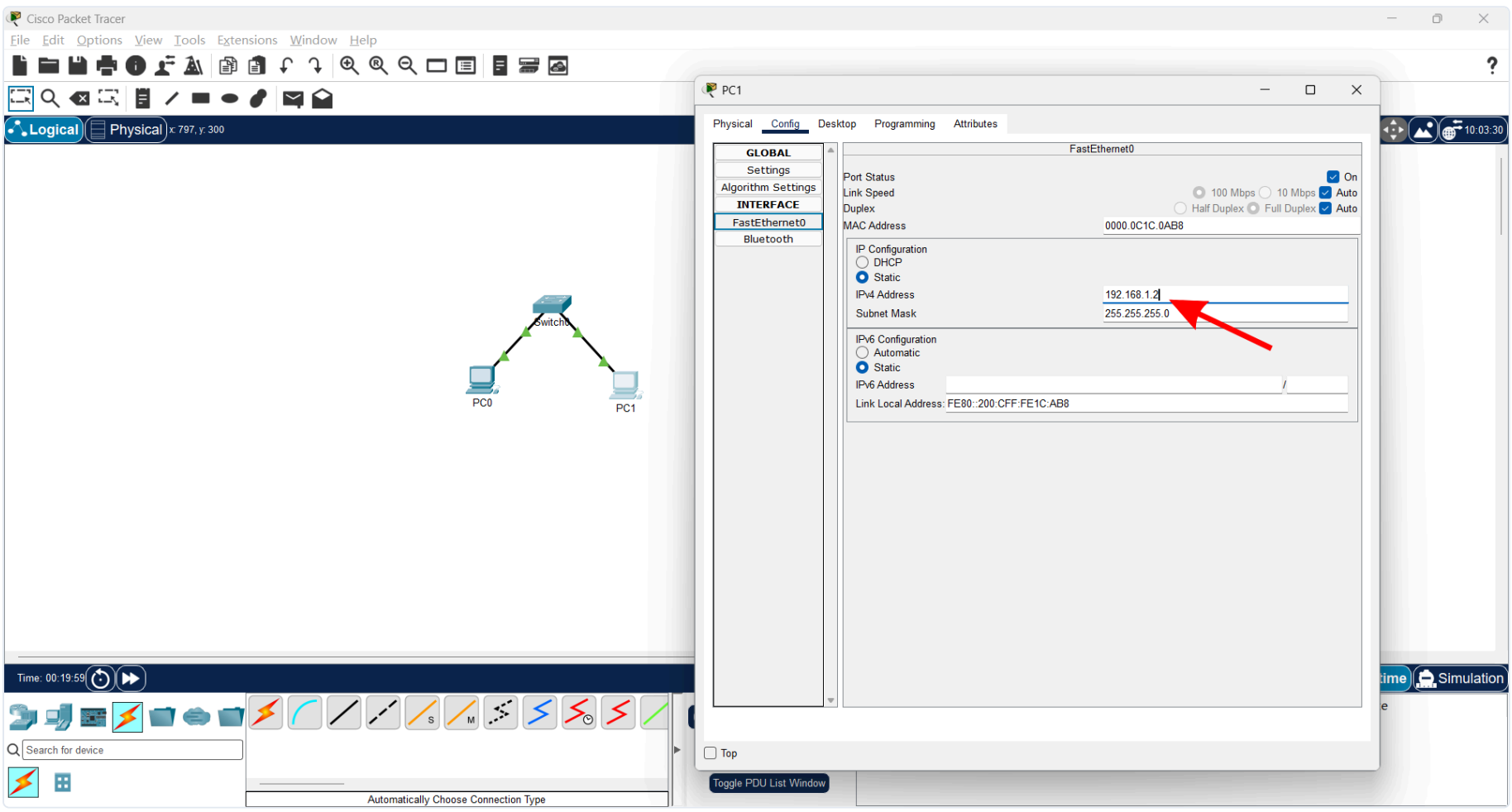
```
ping 192.168.2.1
```

3. Observe and record the test output below (Did the ping succeed? What was the packet loss percentage?):



Step 4: Reconfigure and Verify Direct Communication

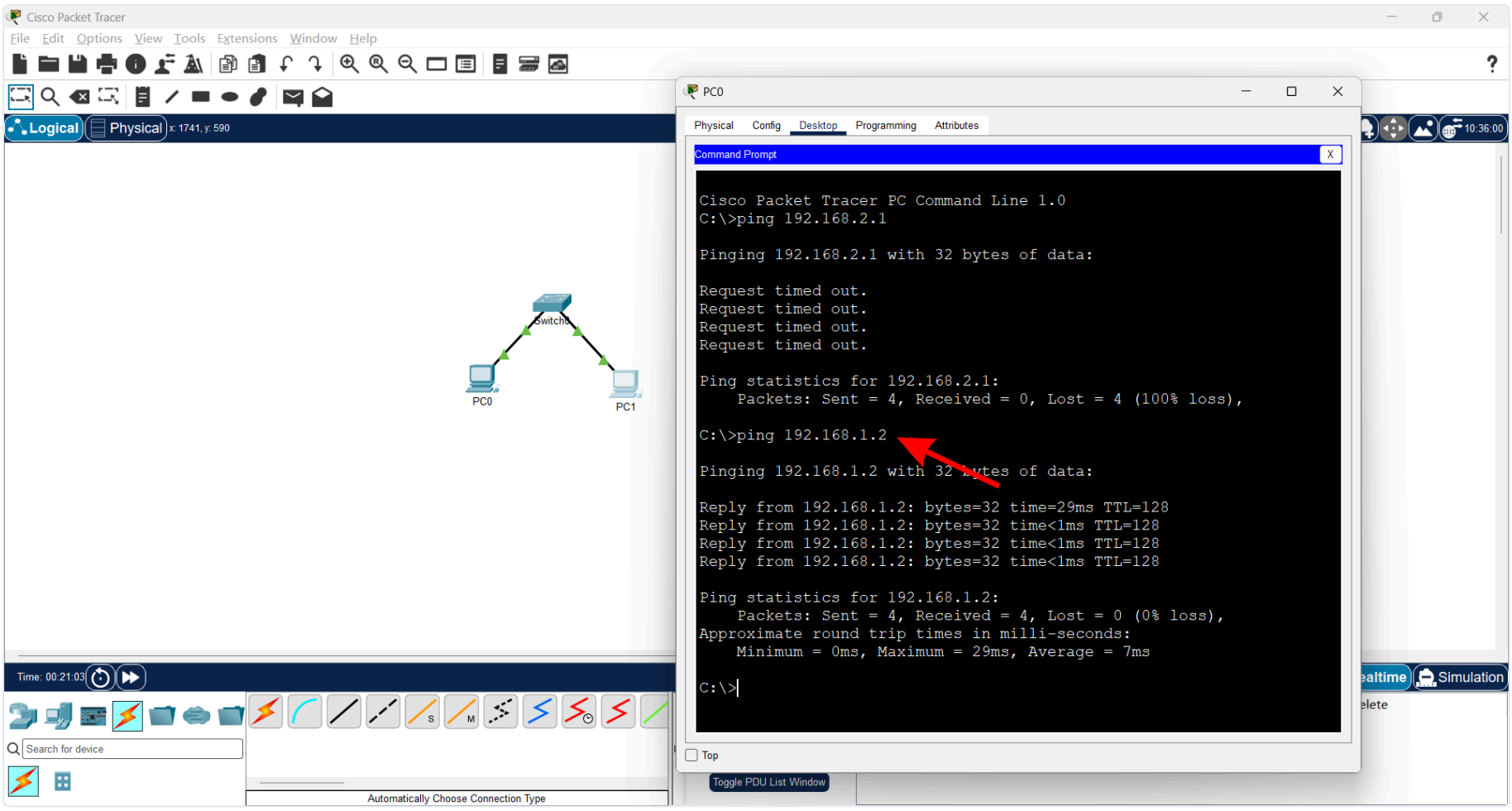
1. Click **PC1** → select the **Config** tab → click **FastEthernet0** (under INTERFACE). Update the IP address to the same network as PC0:
   - IPv4 Address: `192.168.1.2`
   - Subnet Mask: `255.255.255.0`



2. Return to **PC0** Command Prompt and re-test:

```
ping 192.168.1.2
```

3. Verify that the ping succeeds with 0% packet loss.



4. Lab 1 Questions & Analysis

Question 1: Subnet Identification

Why was PC0 unable to communicate with 192.168.2.1 even though both devices were powered on and physically connected to the same switch? Explain using the concept of network IDs defined by the /24 mask.

Question 2: Layer 2 Functionality

Does the switch examine or modify the IP addresses of the packets passing through it? Which address format does the switch inspect to forward data between PC0 and PC1?

Lab 2: Inter-Subnet Communication & Multi-Router Static Routing

1. Scenario & Objectives

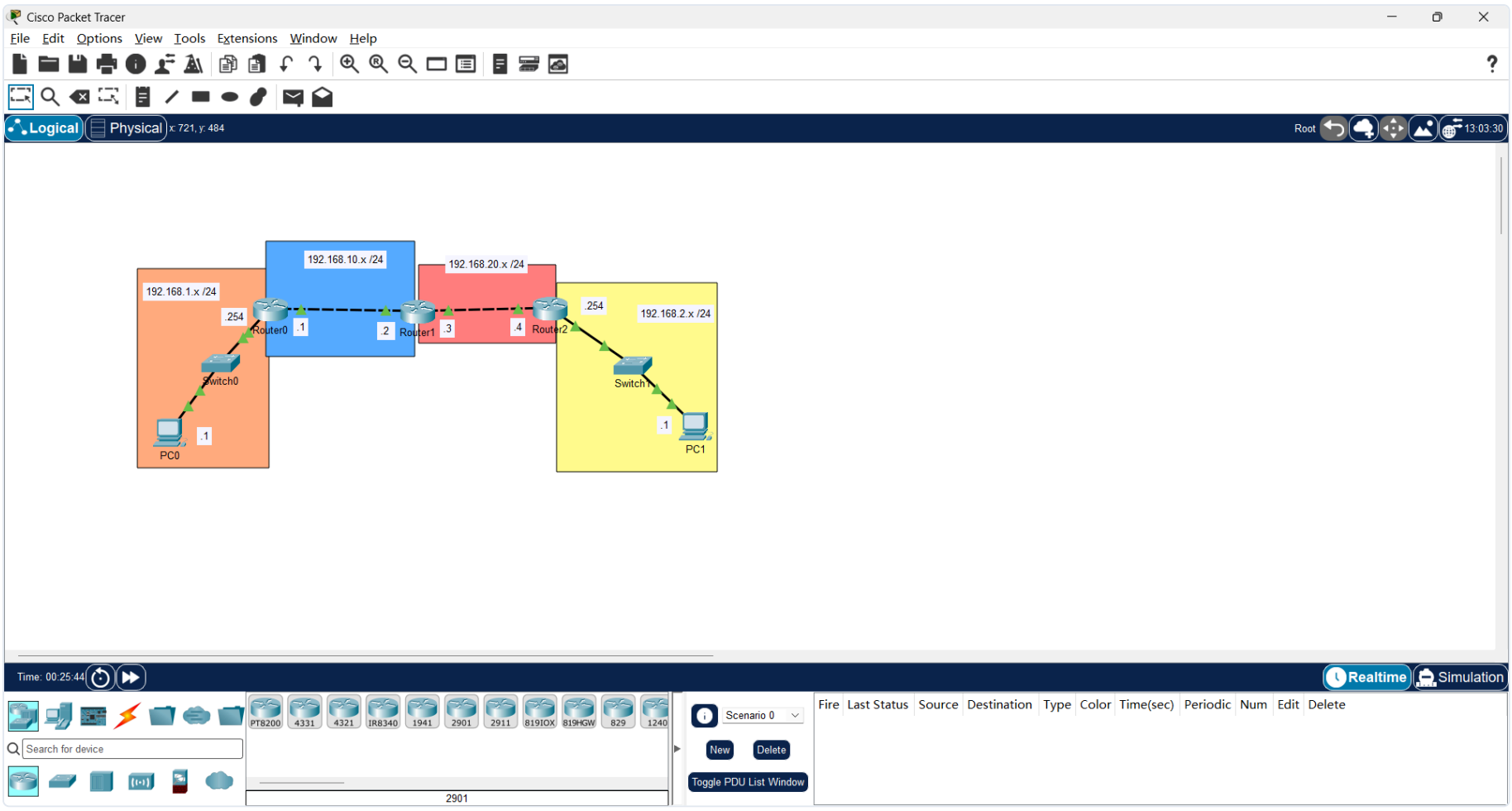
In this lab, you will expand the network across multiple subnets interconnected by three intermediate routers. To achieve end-to-end communication between different subnets, you will configure Default Gateways on endpoint hosts and configure Cisco IOS static routes (`ip route`) on each transit router.

- Configure Default Gateways on endpoint workstations.
- Configure IP addresses on Cisco router GigabitEthernet interfaces and activate links with `no shutdown` .
- Calculate and deploy next-hop static routes for bidirectional reachability.
- Verify end-to-end connectivity across multiple network hops using the `ping` utility.

2. Subnet Architecture & Addressing Plan

> **Network Segmentation Overview (/24 Prefix):**
> • **Network A (Orange Zone):** `192.168.1.0/24` | Gateway: 192.168.1.254

- **Transit Link 1 (Blue Zone):** `192.168.10.0/24` | Route0 ↔ Route1
- **Transit Link 2 (Red Zone):** `192.168.20.0/24` | Route1 ↔ Route2
- **Network B (Yellow Zone):** `192.168.2.0/24` | Gateway: 192.168.2.254

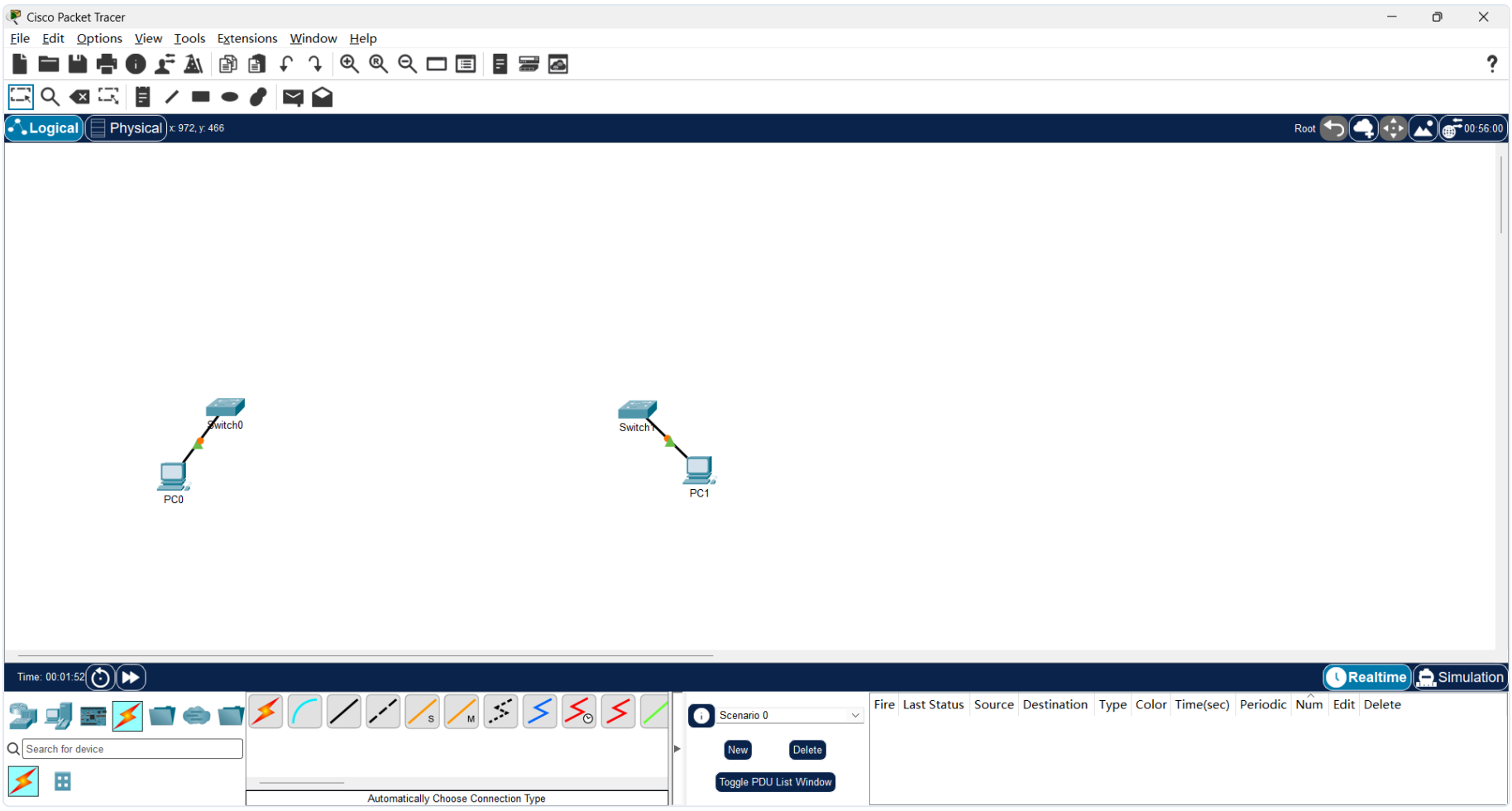


Device	Interface	IP Address	Subnet Mask	Default Gateway / Link Note
PC0	FastEthernet0	192.168.1.1	255.255.255.0	**192.168.1.254**
PC1	FastEthernet0	192.168.2.1	255.255.255.0	**192.168.2.254**
Route0	Gig0/0/0	192.168.1.254	255.255.255.0	LAN Gateway for PC0
	Gig0/0/1	192.168.10.1	255.255.255.0	WAN Link toward Route1
Route1	Gig0/0/0	192.168.10.2	255.255.255.0	WAN Link toward Route0
	Gig0/0/1	192.168.20.3	255.255.255.0	WAN Link toward Route2
Route2	Gig0/0/0	192.168.20.4	255.255.255.0	WAN Link toward Route1
	Gig0/0/1	192.168.2.254	255.255.255.0	LAN Gateway for PC1

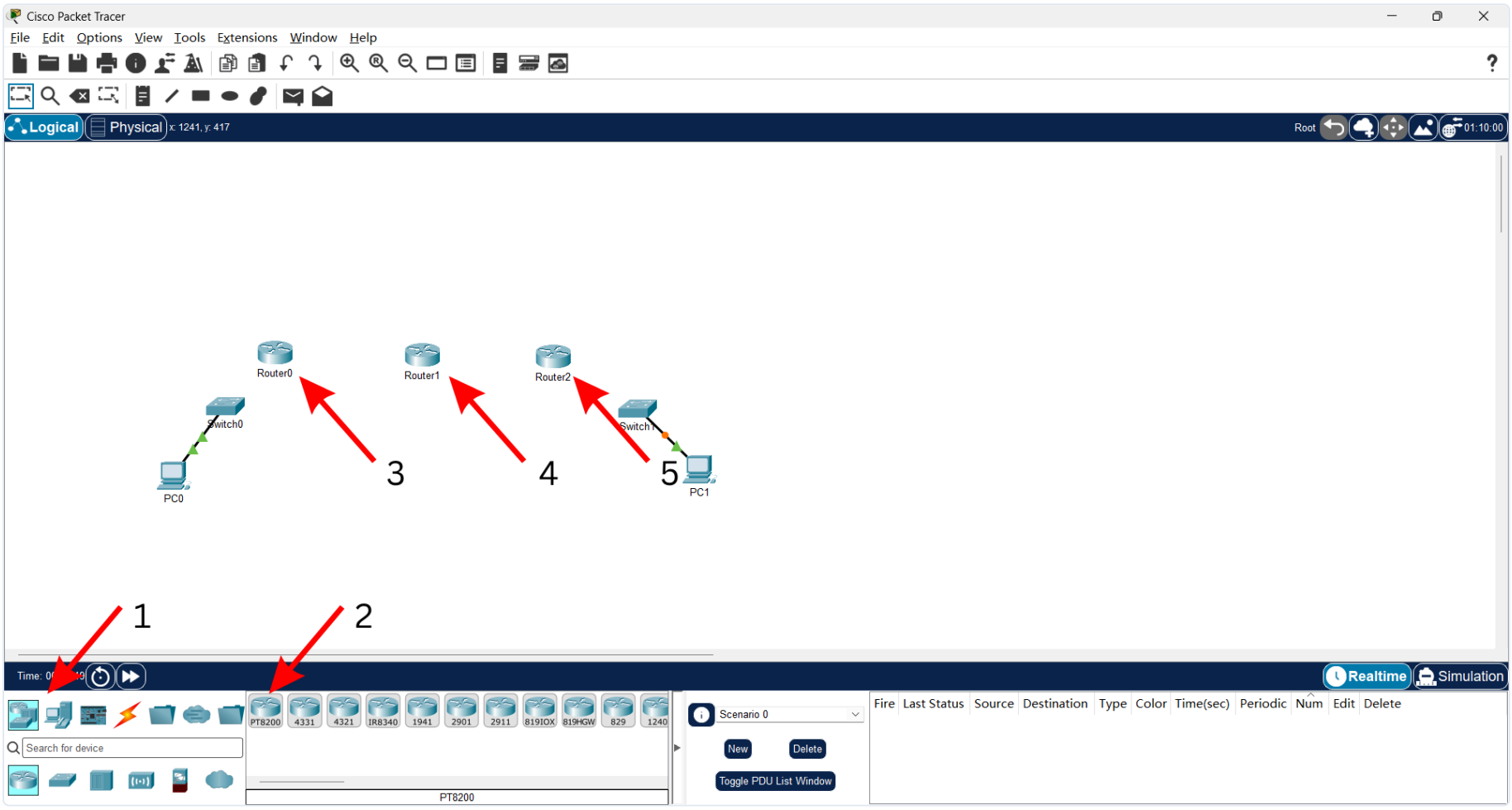
3. Step-by-Step Procedure

Step 1: Construct the Multi-Router Topology

1. Deploy 2 PCs and 2 switches (2960).

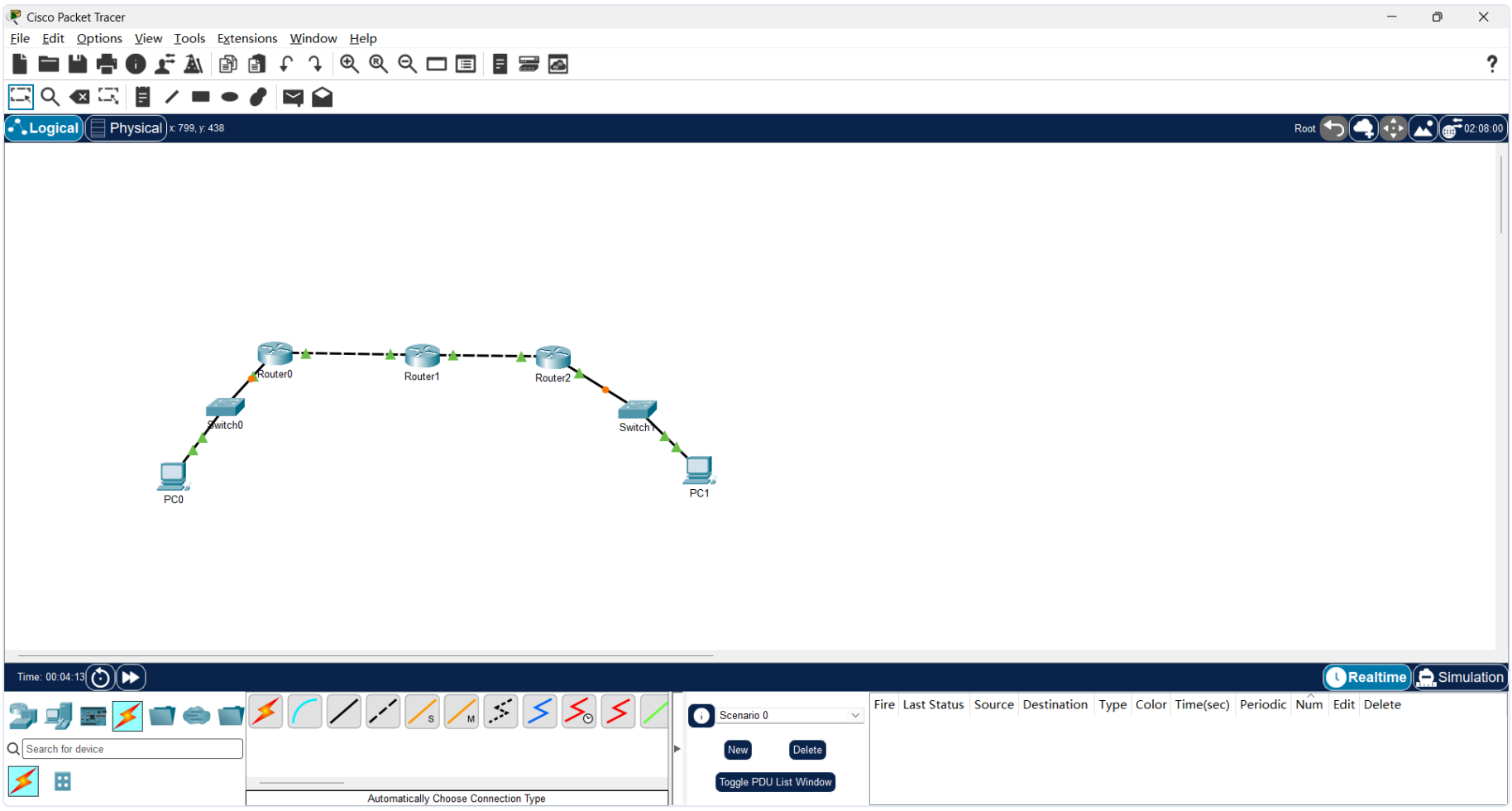


2. Deploy 3 routers (Route0, Route1, Route2).



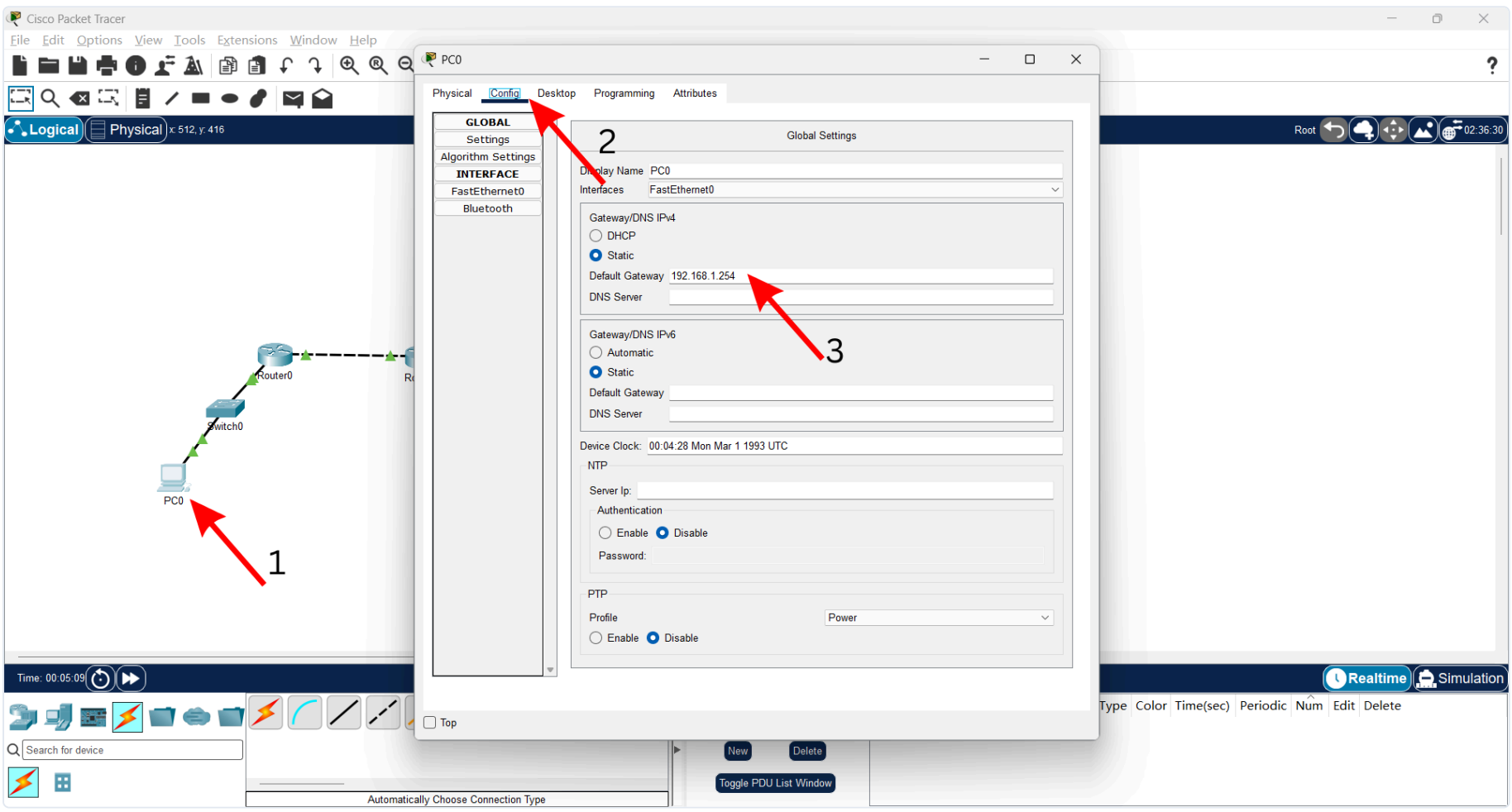
3. Connect the devices linearly using connection cables:

```
PC0 → Switch0 → Route0 → Route1 → Route2 → Switch1 → PC1
```

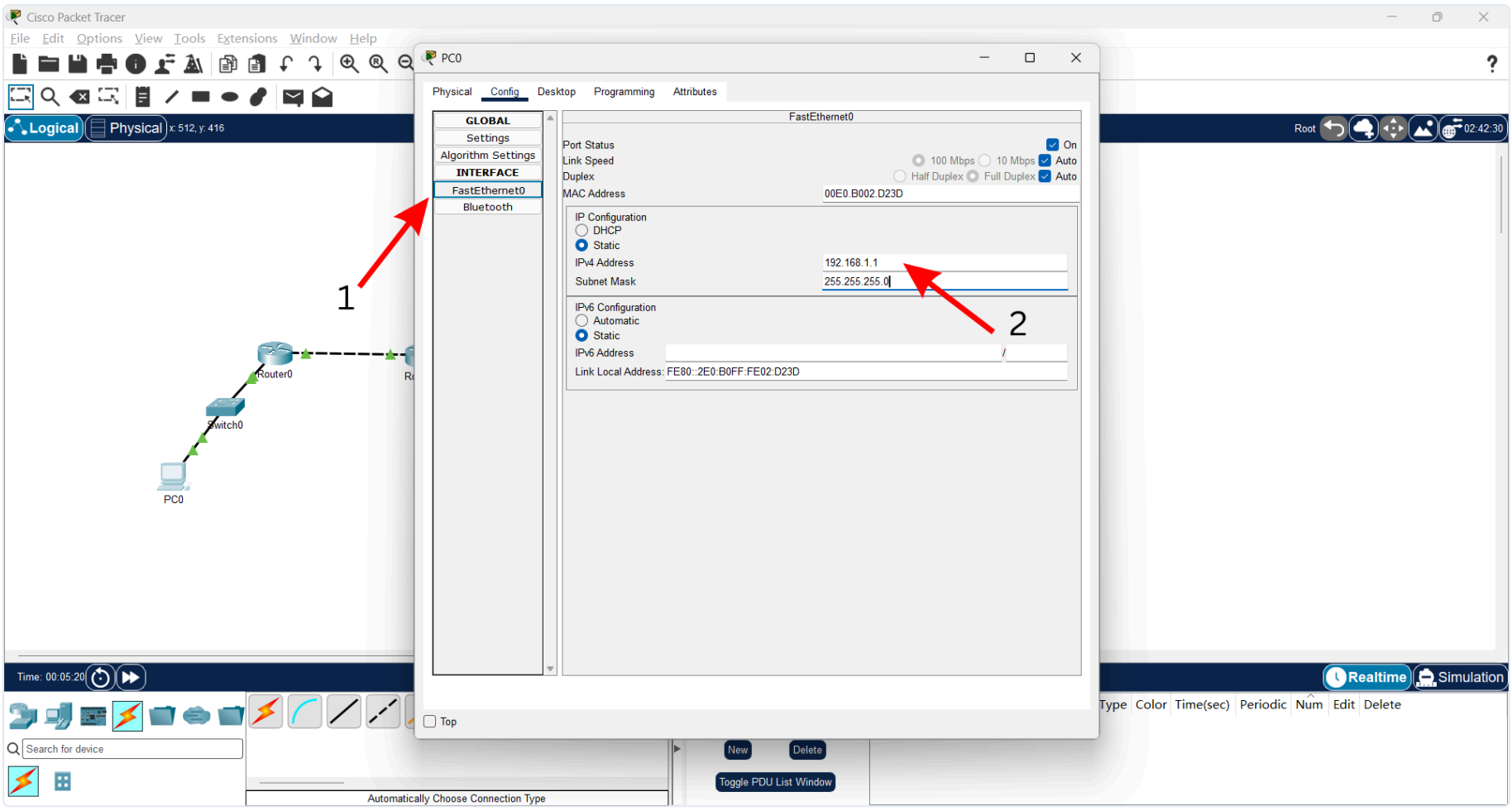


Step 2: Assign Host IP Settings and Default Gateways

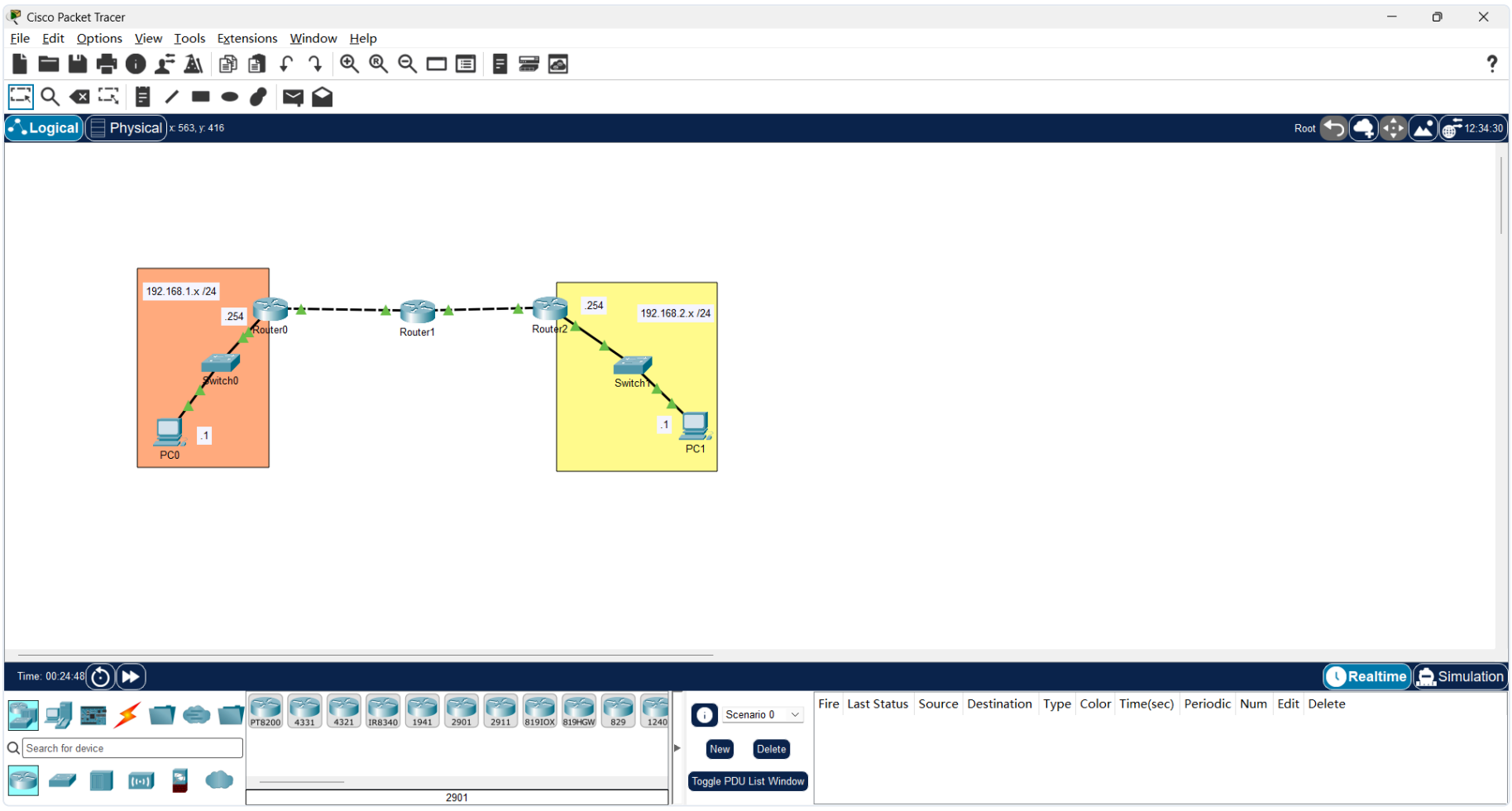
1. Click **PC0** → select the **Config** tab → click **Global Settings**:
   - Default Gateway: `192.168.1.254`



2. Click **FastEthernet0** (under INTERFACE on PC0):
   - IPv4 Address: `192.168.1.1`
   - Subnet Mask: `255.255.255.0`



3. Configure **PC1** with corresponding settings:
    - Default Gateway: `192.168.2.254`
    - IPv4 Address: `192.168.2.1`
    - Subnet Mask: `255.255.255.0`



Step 3: Configure Router Interfaces

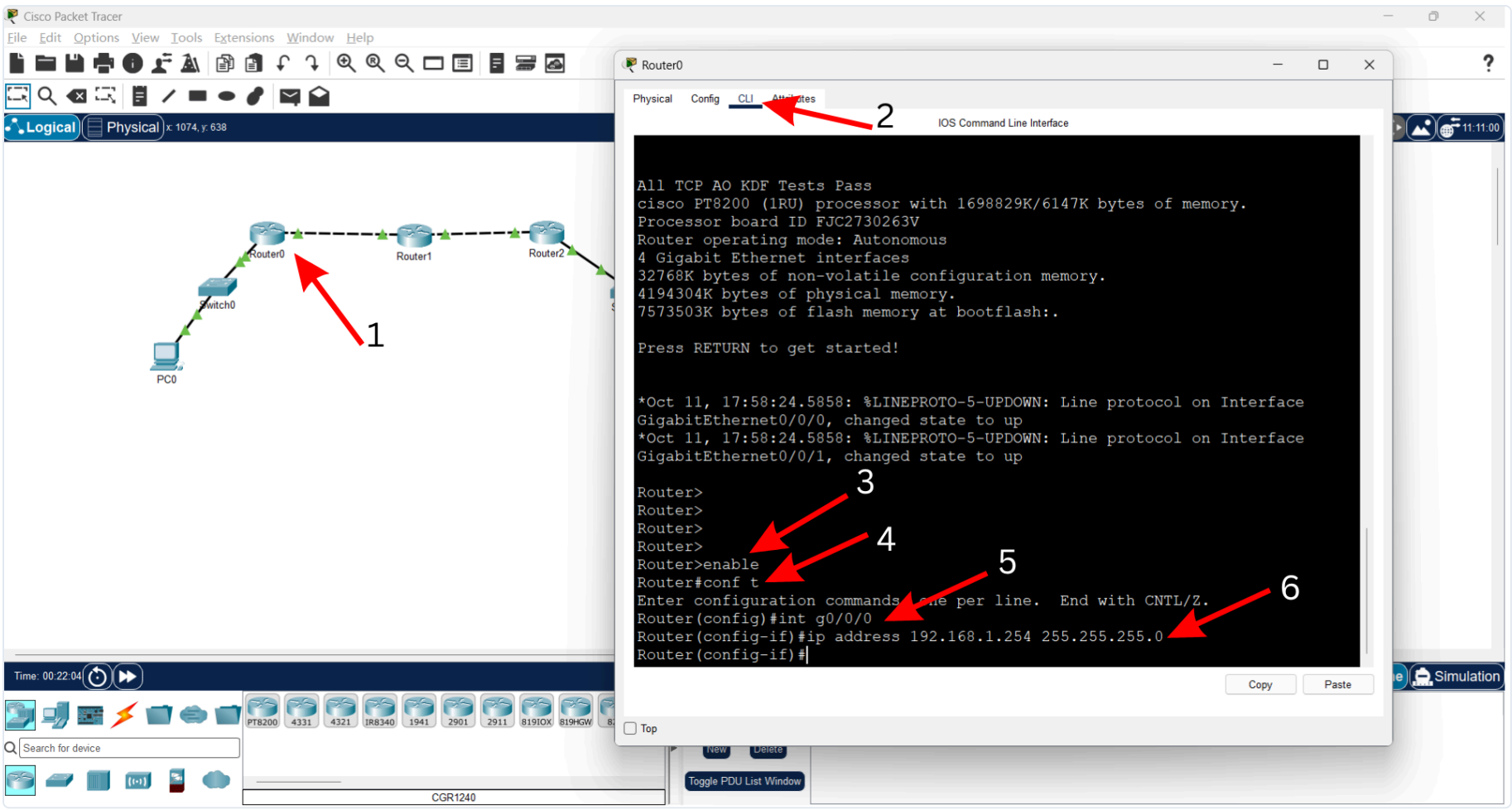
> **Important Notice:** Cisco router physical interfaces are disabled by default. You must enter `no shutdown` under each interface configuration mode to activate the link.

1. Configure Route0: Click Route0 → select the **CLI** tab:

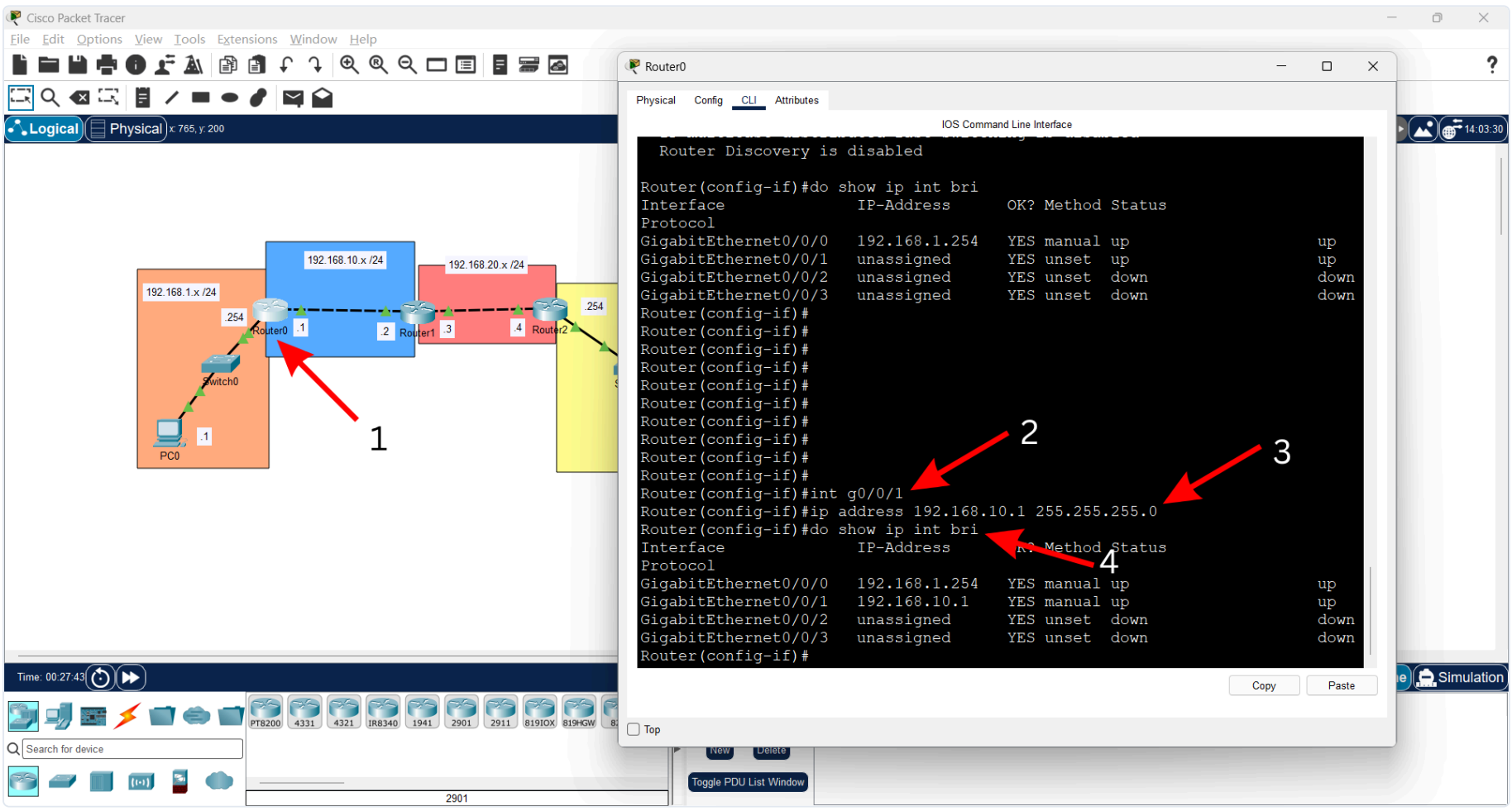
```
! Step A: Enter Configuration Mode
Router> enable
Router# configure terminal

! Step B: Configure the LAN Gateway Interface for PC0
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.1.254 255.255.255.0
```

```
Router(config-if)# no shutdown
Router(config-if)# exit
```




```
! Step C: Configure the WAN Interface toward Route1
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

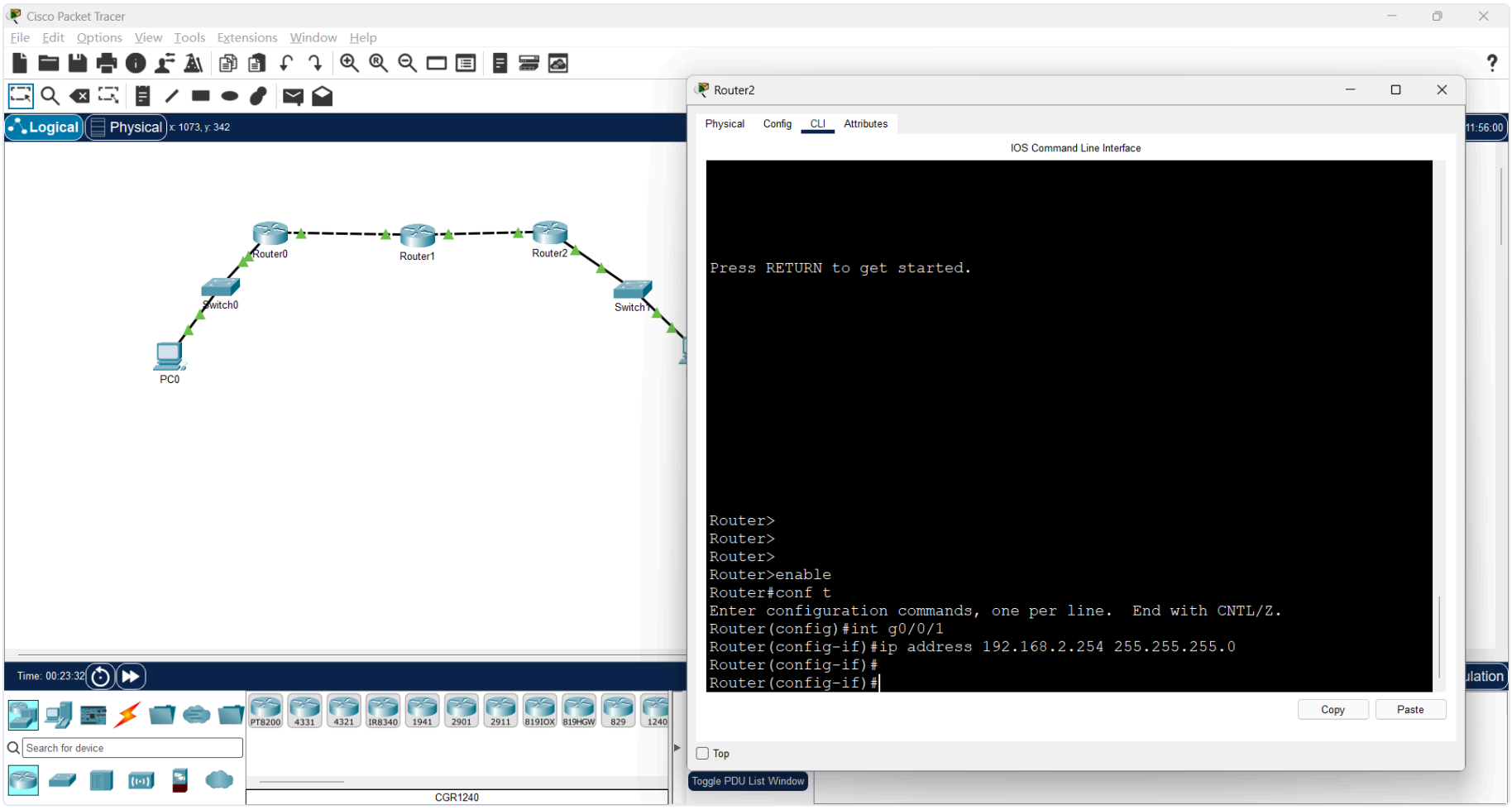


2. Configure Route2: Open the CLI tab on Route2:

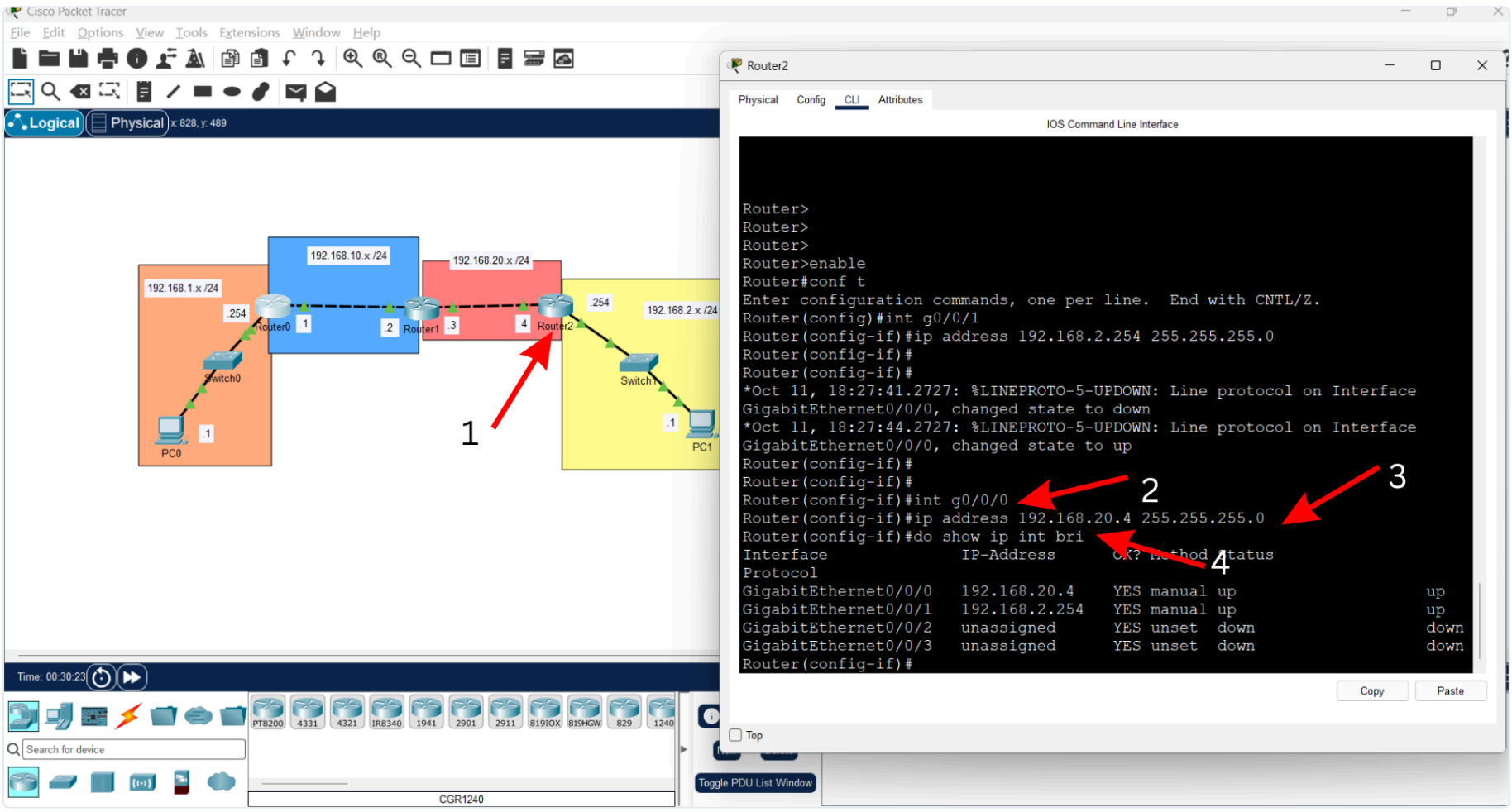
```
! Step A: Enter Configuration Mode
Router> enable
Router# configure terminal

! Step B: Configure the LAN Gateway Interface for PC1
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.2.254 255.255.255.0
```

```
Router(config-if)# no shutdown
Router(config-if)# exit
```




```
! Step C: Configure the WAN Interface toward Route1
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.20.4 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```



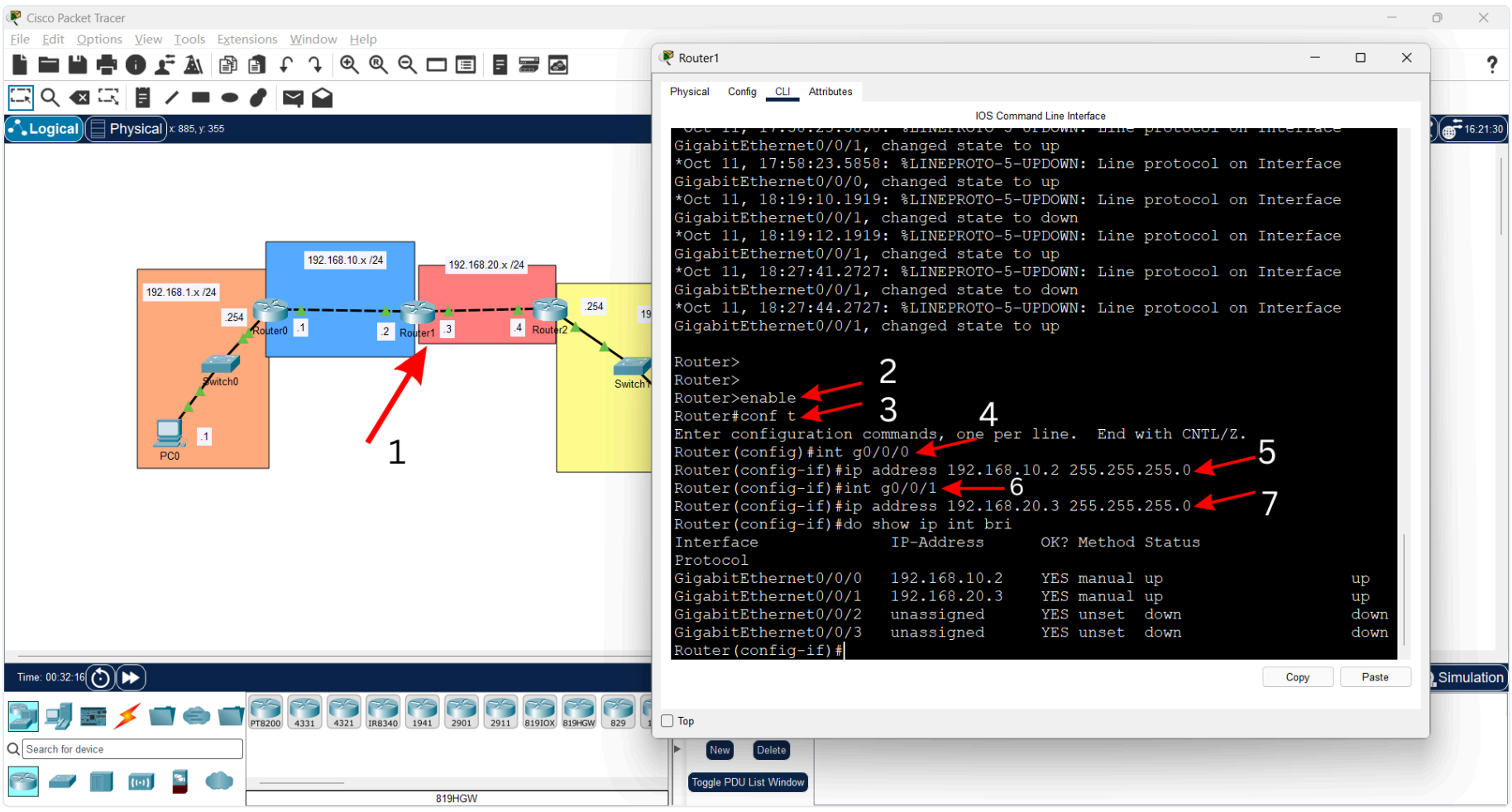
3. Configure Transit Router Route1: Open the CLI tab on Route1:

```
! Step A: Enter Configuration Mode
Router> enable
Router# configure terminal

! Step B: Configure Interface toward Route0
Router(config)# interface gigabitethernet 0/0/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
```

```
Router(config-if)# exit

! Step C: Configure Interface toward Route2
Router(config)# interface gigabitethernet 0/0/1
Router(config-if)# ip address 192.168.20.3 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

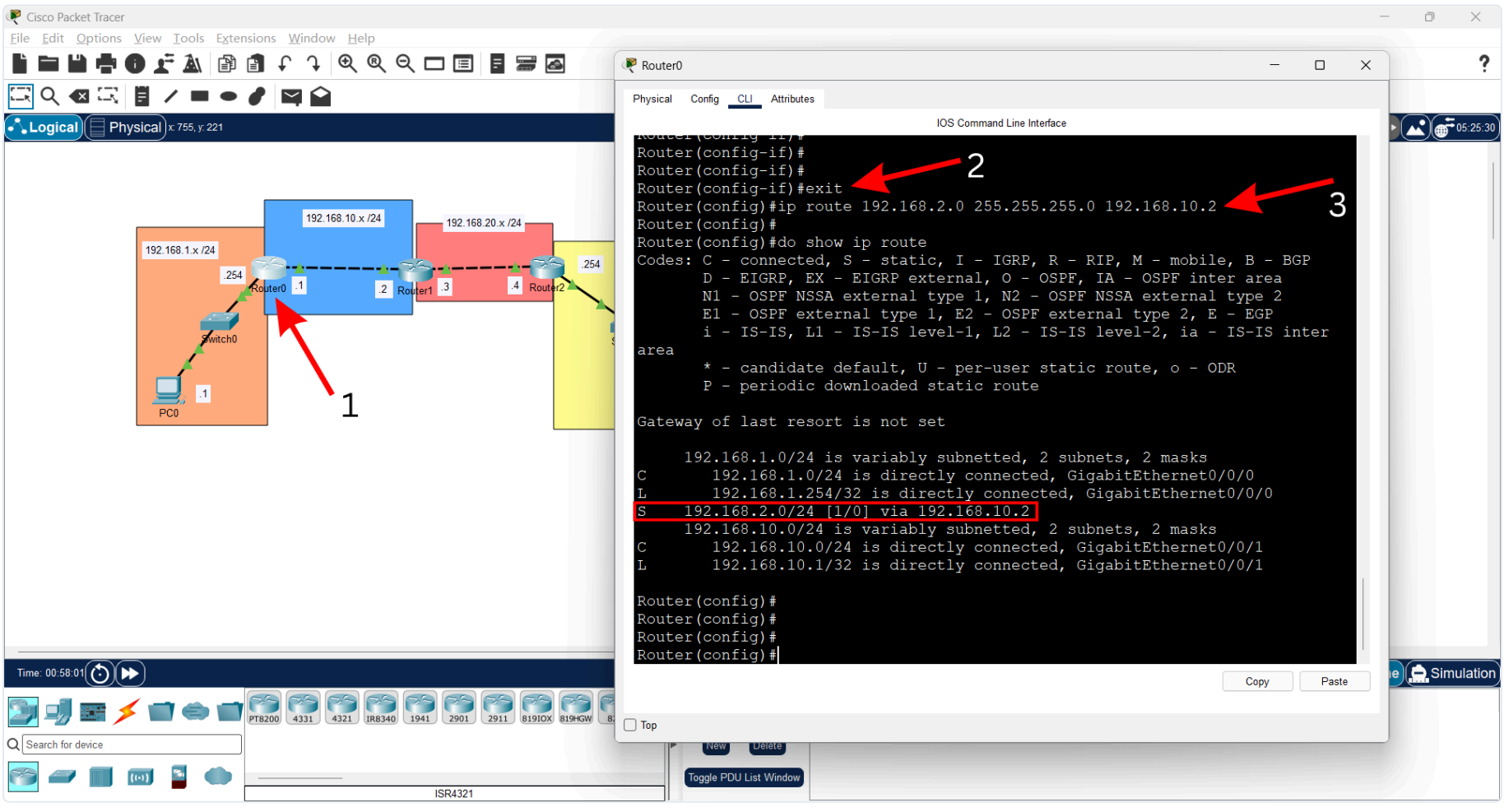


Step 4: Configure Static Routes

Static route command syntax: `ip route [Destination_Network] [Subnet_Mask] [Next_Hop_IP]`

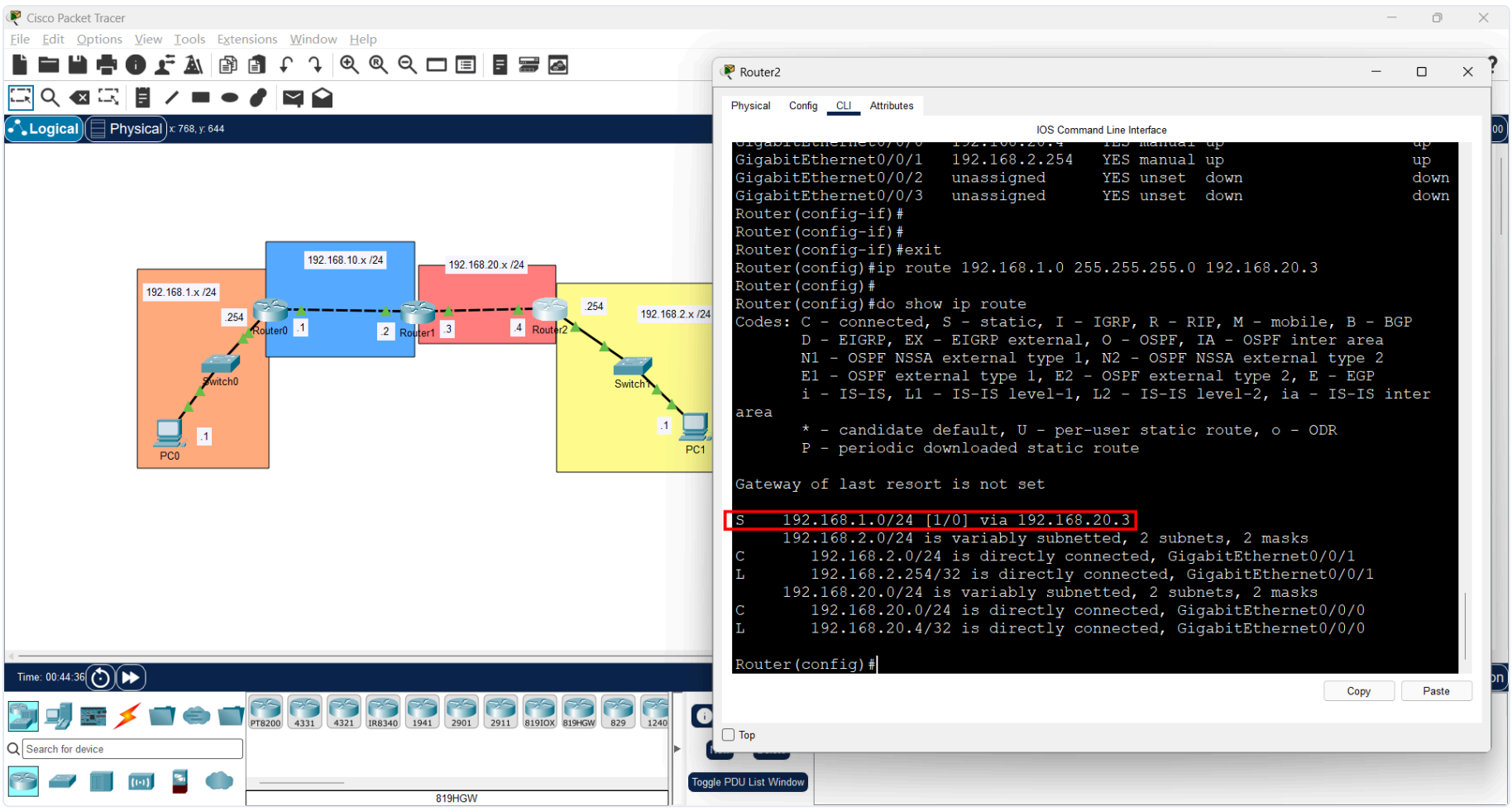
1. On **Route0**, add the route toward Network B (192.168.2.0/24):

```
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.10.2
Router(config)# do show ip route
```



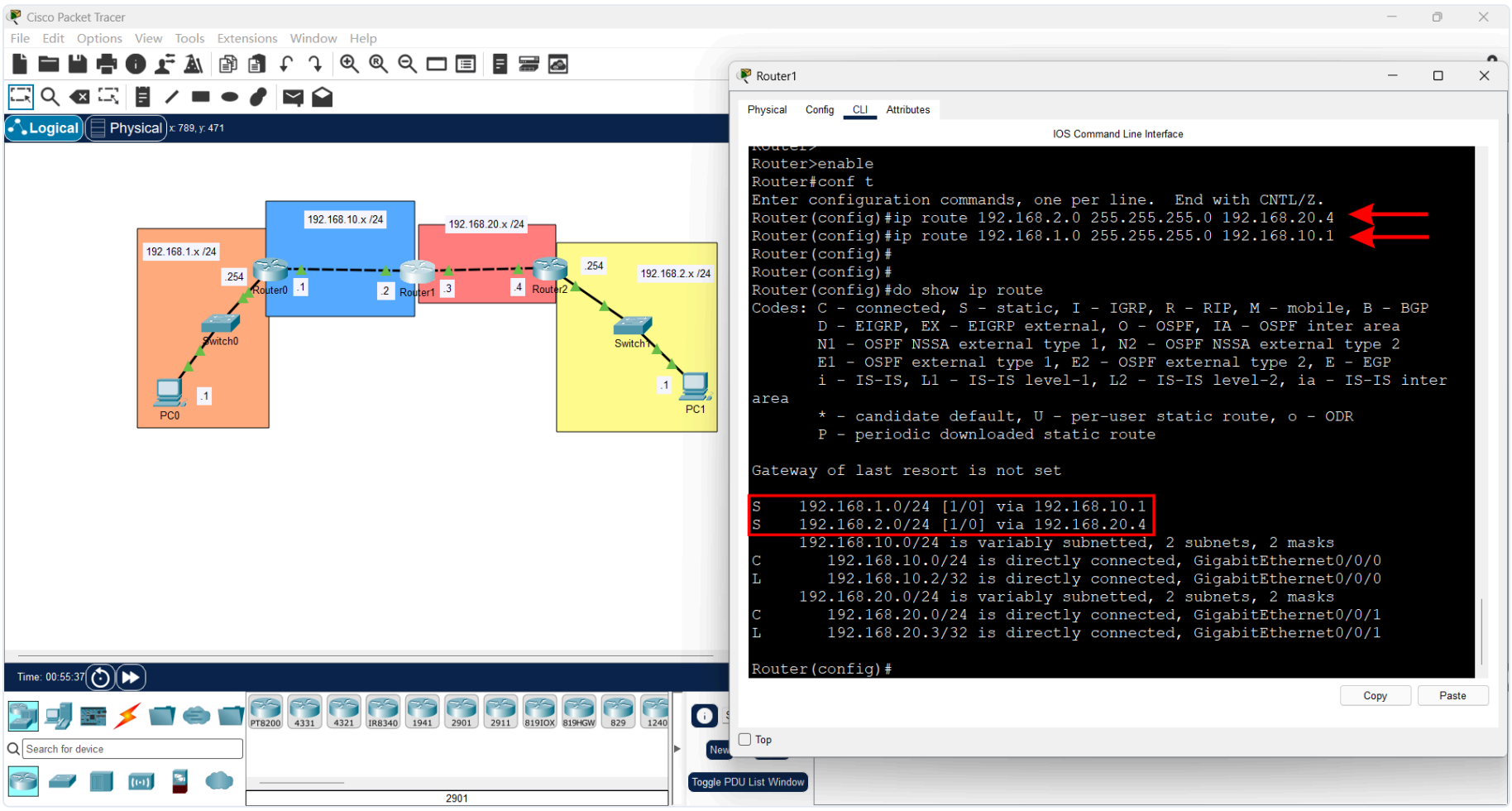
2. On **Route2**, add the reverse route toward Network A (192.168.1.0/24):

```
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.20.3
Router(config)# do show ip route
```



3. On **Route1**, add routes in both directions:

```
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.20.4
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)# do show ip route
```

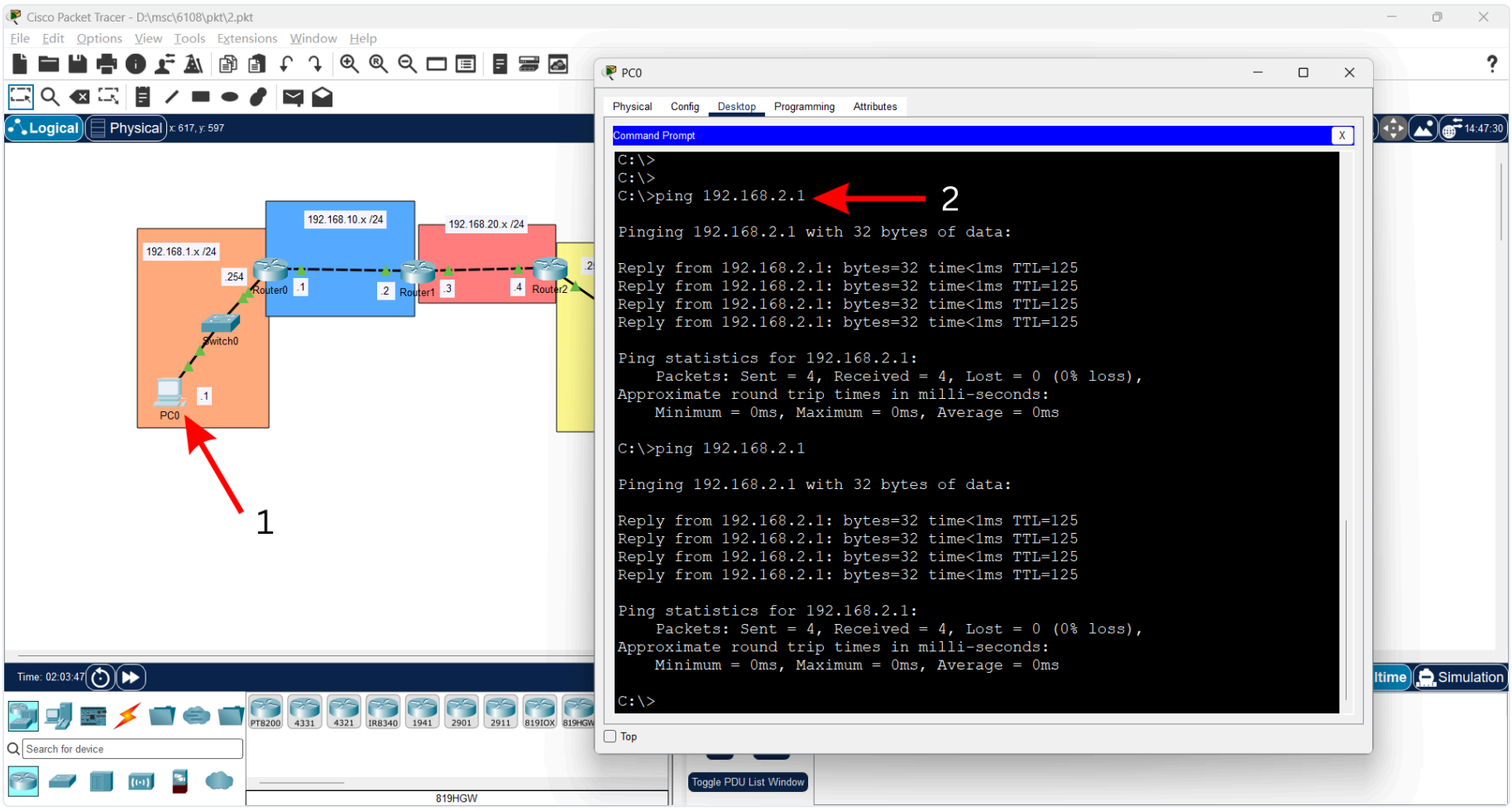


Step 5: End-to-End Connectivity Verification

1. Open **PC0** → Command Prompt.
2. Execute:

```
ping 192.168.2.1
```

3. Verify that the ping succeeds with 0% packet loss.



4. Lab 2 Questions & Analysis

Question 1: Default Gateway Function

Why must PC0 configure a Default Gateway (192.168.1.254) in this lab to reach PC1, whereas in Lab 1 no Default Gateway was required? Explain how host routing behavior changes when the destination IP is on a different subnet.

Question 2: Asymmetric Routing Test

If Route0 and Route1 are configured properly, but the return route command on Route2 is omitted, what error will PC0 display when pinging PC1? Why?

Question 3: Missing Intermediate Route

What would happen if the static routes on intermediate router Route1 were completely missing? Would PC0 receive an ICMP Echo Reply from PC1? Explain the forwarding failure point.

Lab 3: VLAN Segmentation & 802.1Q Trunking

1. Scenario & Objectives

In this lab, you will configure Virtual Local Area Networks (VLANs) across two interconnected Layer 2 switches. You will observe how VLANs create isolated broadcast domains within the same physical infrastructure and configure an 802.1Q Trunk link to carry multiple VLAN traffic between switches.

- Create VLAN 10 and VLAN 20 on two Cisco Catalyst 2960 switches.

- Assign switch access ports to respective VLANs for host isolation.
- Configure the inter-switch link as an 802.1Q Trunk port allowing VLANs 10 and 20.
- Verify that hosts in the same VLAN communicate across switches while cross-VLAN traffic is blocked.

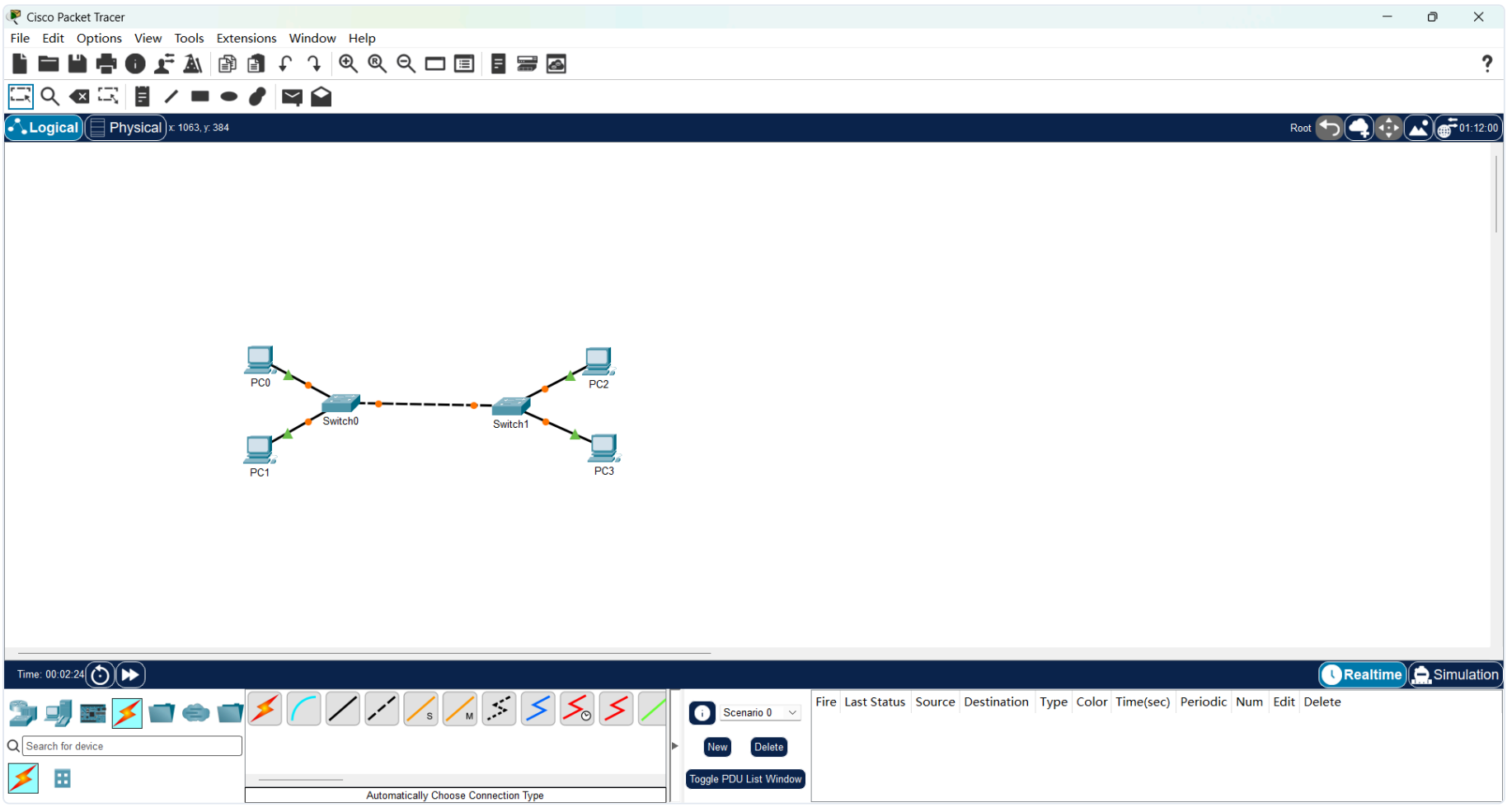
2. Addressing & Port Assignment Table

Device	Interface	IP Address	Subnet Mask	VLAN Assignment	Connected Switch Port
PC0	FastEthernet0	192.168.1.1	255.255.255.0	VLAN 10	Switch0 Fa0/1
PC1	FastEthernet0	192.168.1.2	255.255.255.0	VLAN 20	Switch0 Fa0/2
PC2	FastEthernet0	192.168.1.3	255.255.255.0	VLAN 10	Switch1 Fa0/2
PC3	FastEthernet0	192.168.1.4	255.255.255.0	VLAN 20	Switch1 Fa0/3
Trunk Link	Fa0/3 ↔ Fa0/1	N/A	N/A	VLANs 10, 20 Allowed	Switch0 Fa0/3 ↔ Switch1 Fa0/1

3. Step-by-Step Procedure

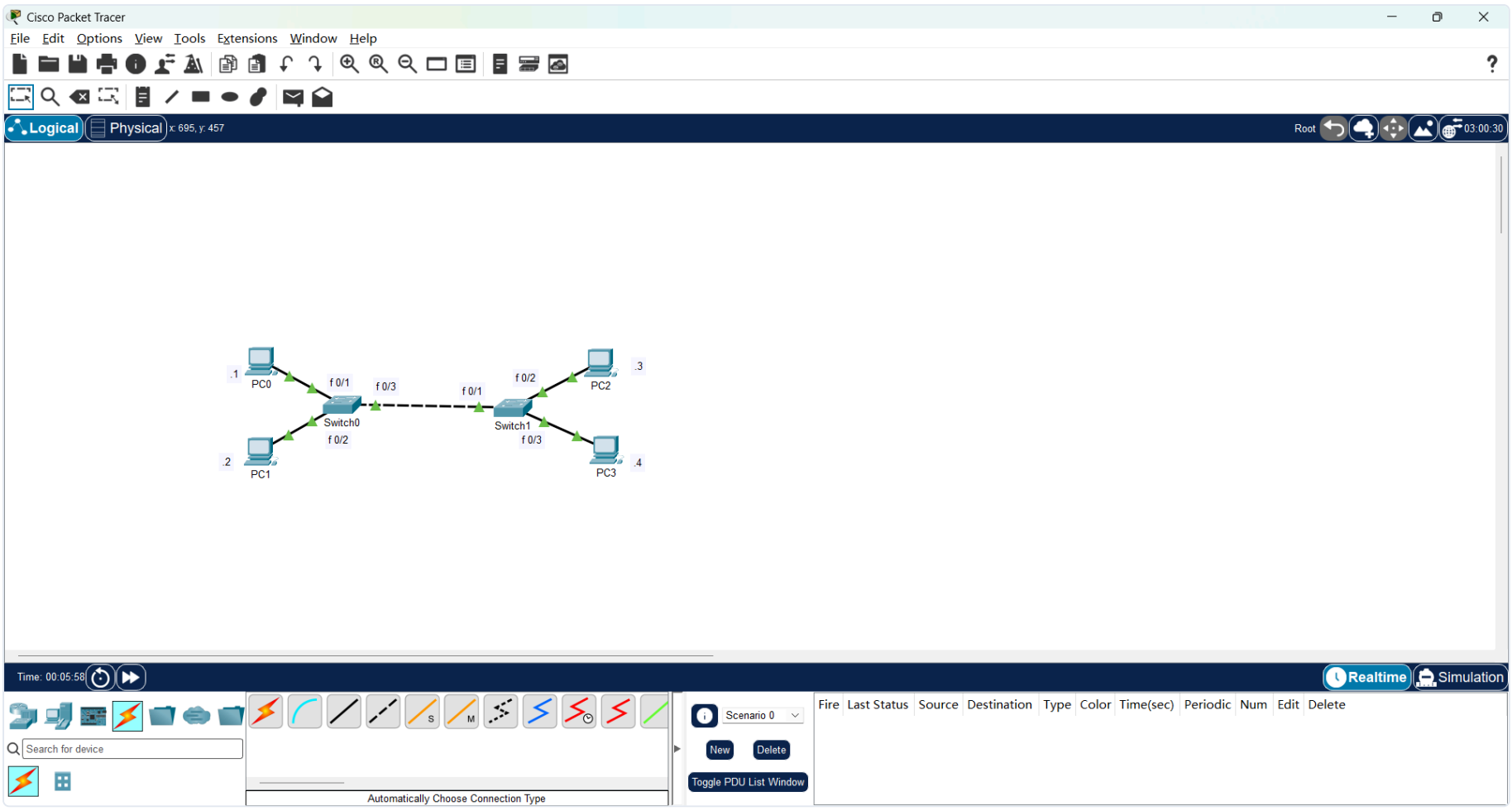
Step 1: Place Hardware Components and Connect Cables

1. Deploy 4 PCs (PC0, PC1, PC2, PC3) and 2 Cisco Catalyst 2960 switches (Switch0, Switch1).
2. Use **Automatically Choose Connection Type** (lightning bolt icon) to connect:
   - PC0 to Switch0 (Fa0/1), PC1 to Switch0 (Fa0/2).
   - PC2 to Switch1 (Fa0/2), PC3 to Switch1 (Fa0/3).
   - Switch0 (Fa0/3) to Switch1 (Fa0/1).



Step 2: Assign Static IP Addresses on All PCs

1. On each PC, click the **Config** tab → **FastEthernet0** (under INTERFACE) and assign IP addresses according to the addressing table:
   - PC0: `192.168.1.1` / `255.255.255.0`
   - PC1: `192.168.1.2` / `255.255.255.0`
   - PC2: `192.168.1.3` / `255.255.255.0`
   - PC3: `192.168.1.4` / `255.255.255.0`

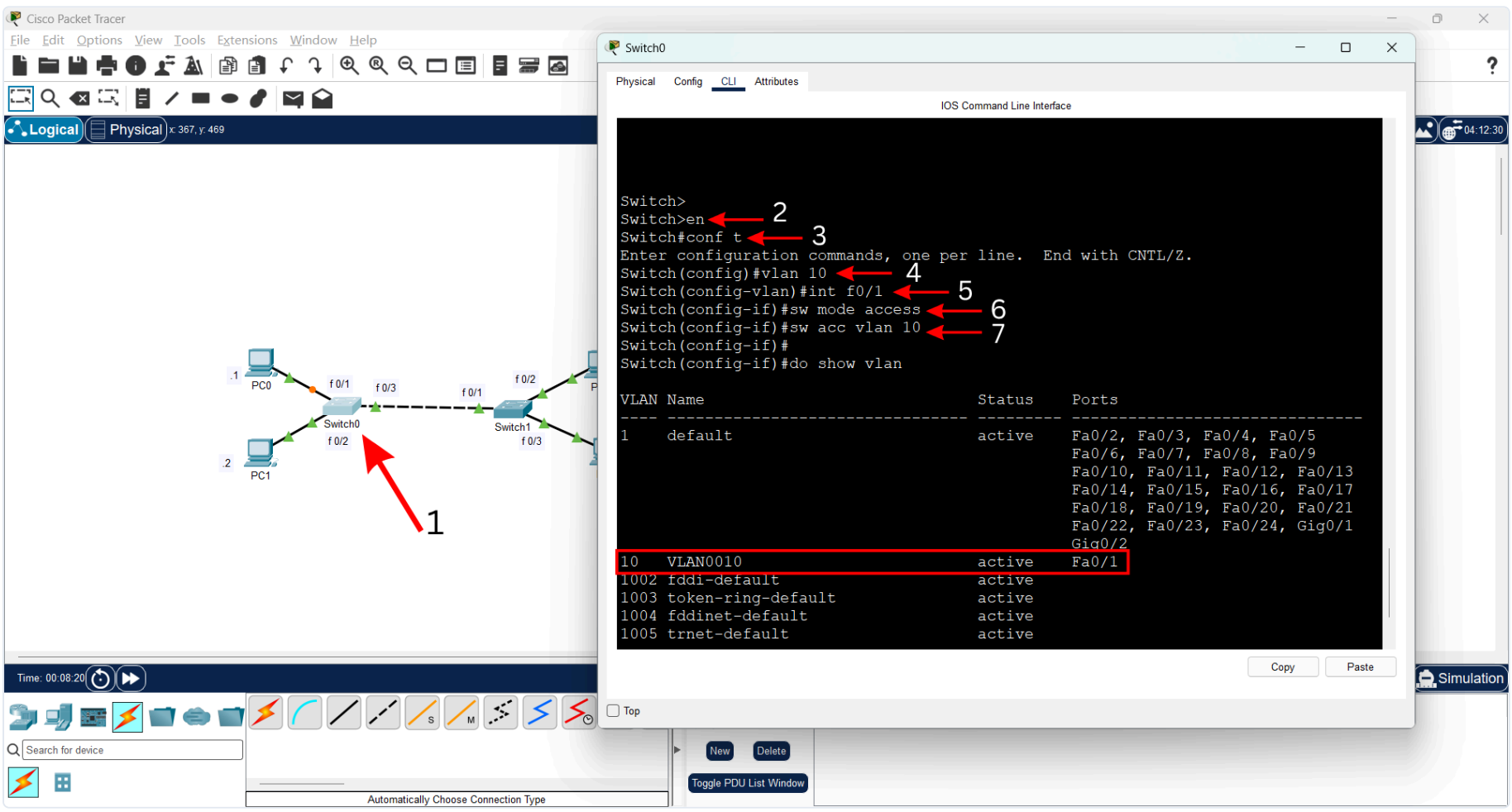


Step 3: Configure Access VLANs on Switch0

1. Click **Switch0** → open the **CLI** tab:

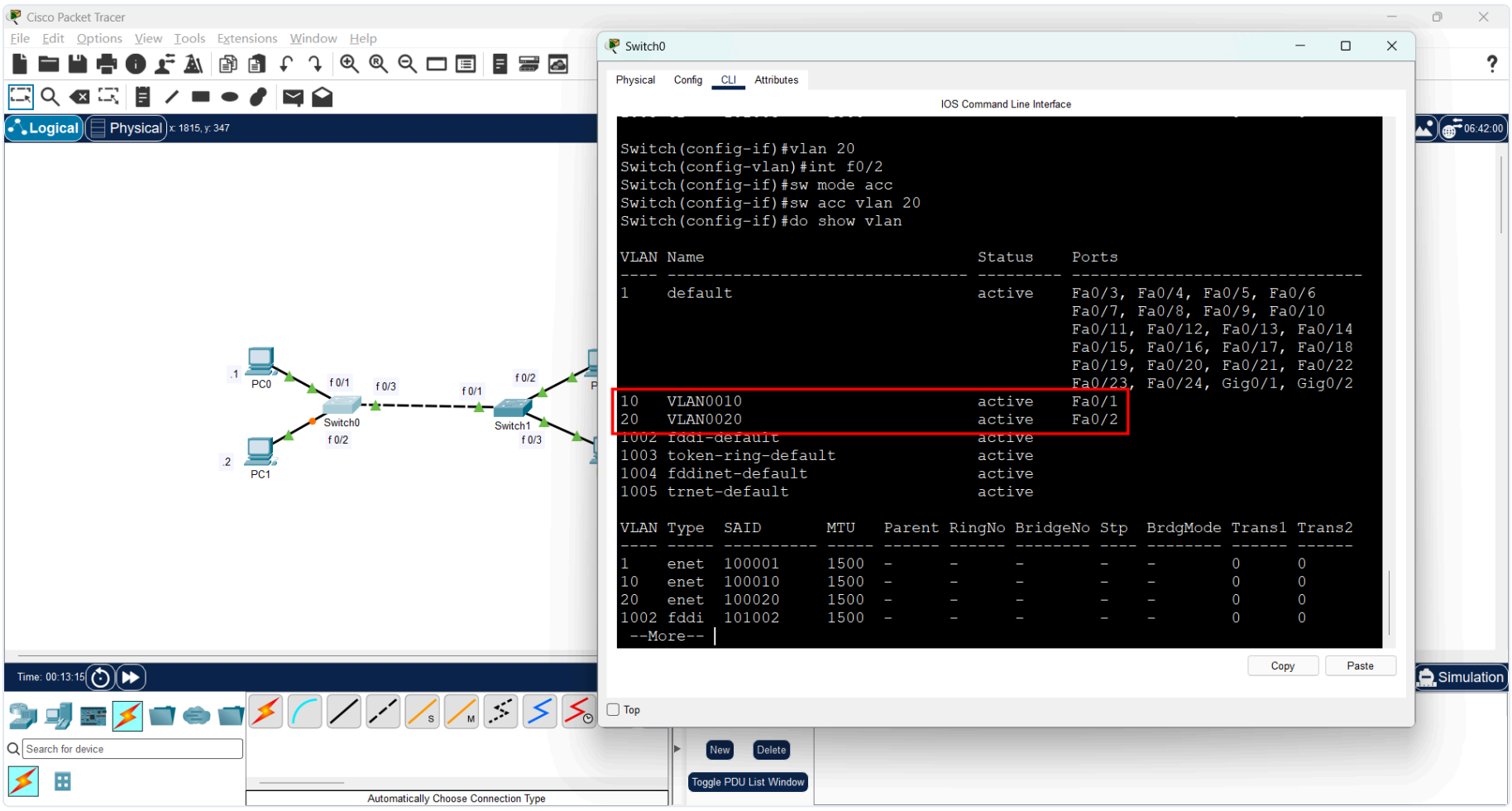
```
Switch> enable
Switch# configure terminal

! Create VLAN 10 and assign Fa0/1
Switch(config)# vlan 10
Switch(config-vlan)# exit
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
```




```
! Create VLAN 20 and assign Fa0/2
Switch(config)# vlan 20
Switch(config-vlan)# exit
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
Switch(config-if)# do show vlan
```



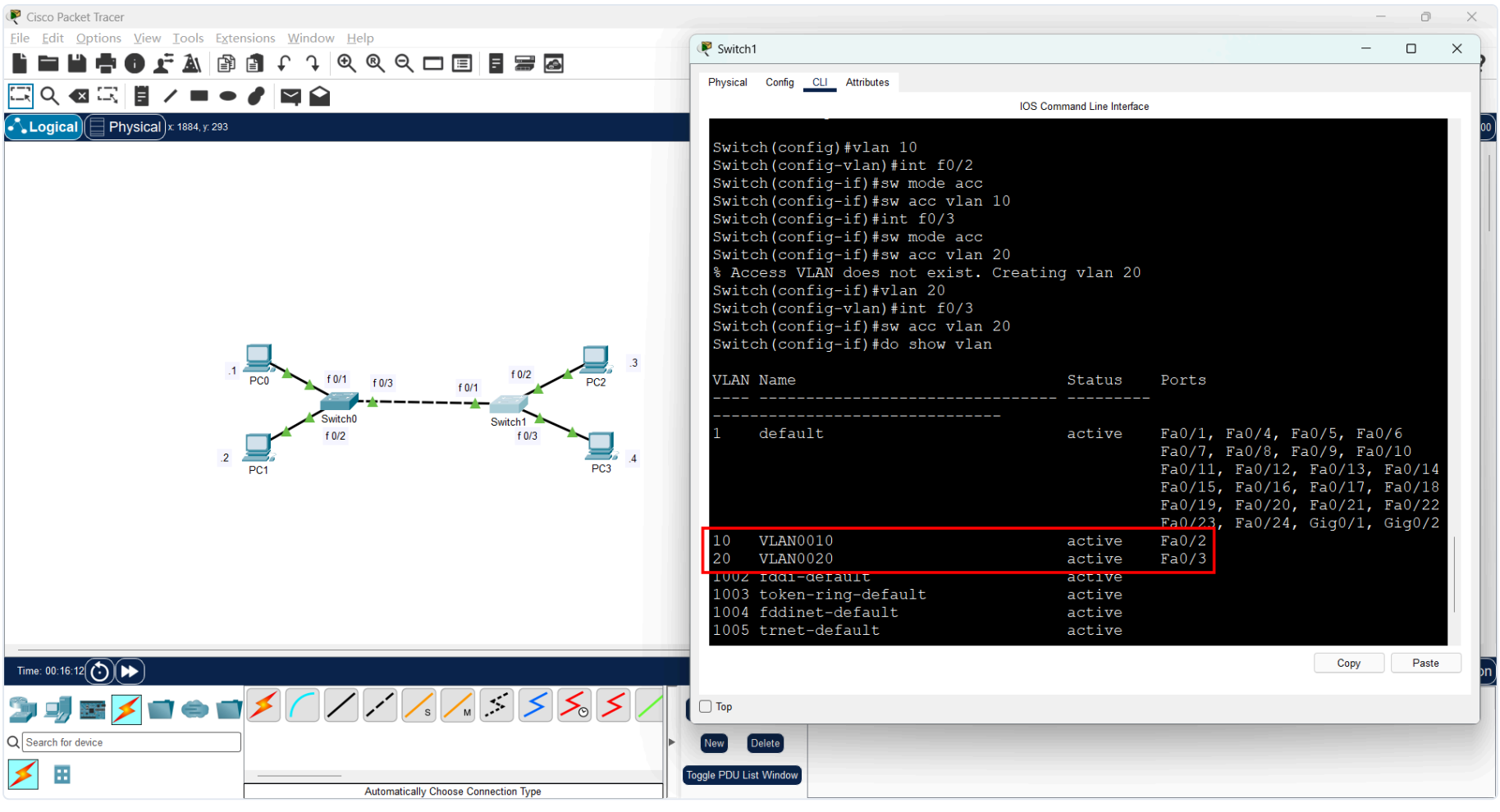
Step 4: Configure Access VLANs on Switch1

1. Click **Switch1** → open the **CLI** tab:

```
Switch> enable
Switch# configure terminal

! Create VLAN 10 and assign Fa0/2
Switch(config)# vlan 10
Switch(config)# interface fastethernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

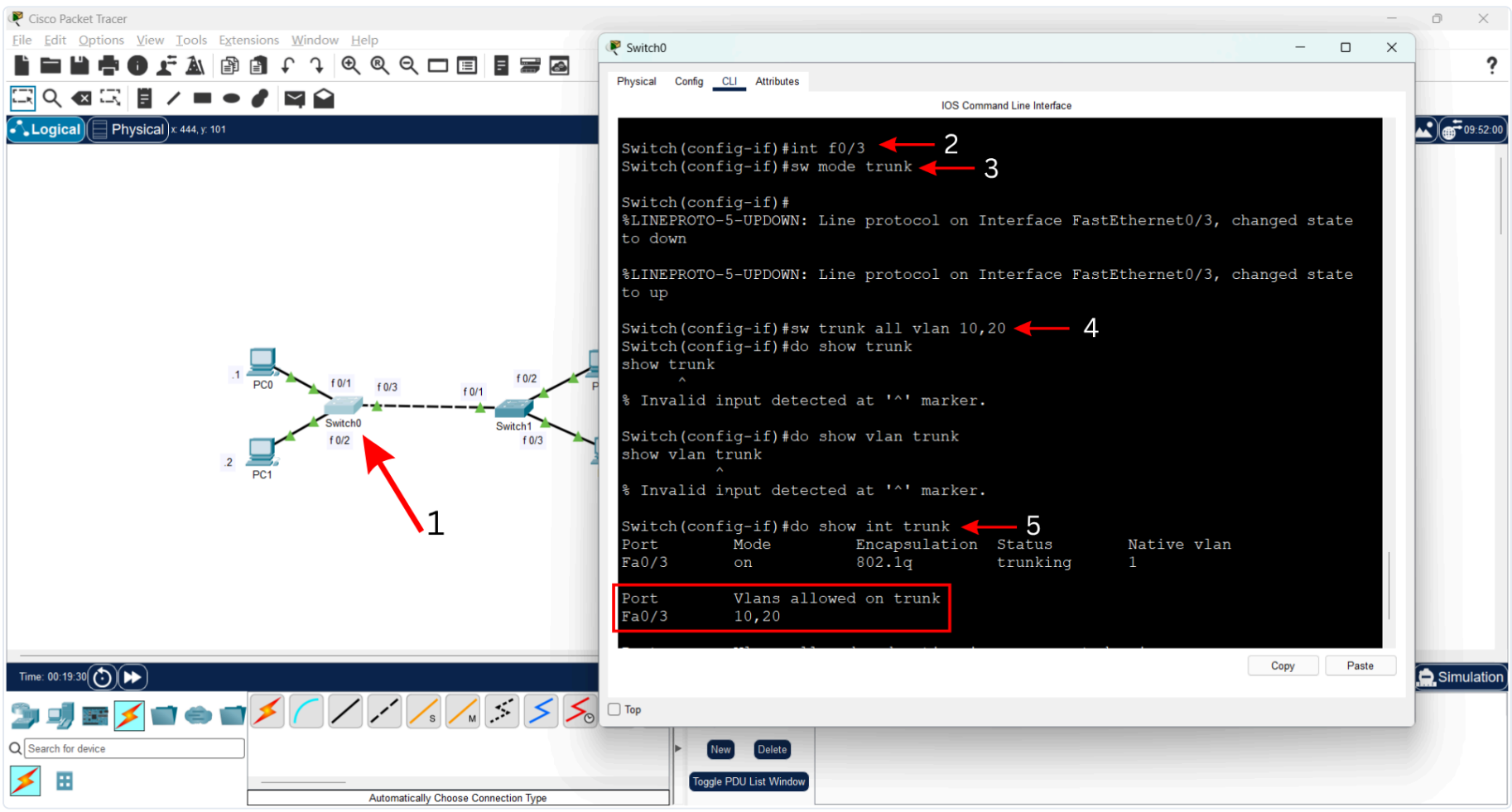
! Create VLAN 20 and assign Fa0/3
Switch(config)# vlan 20
Switch(config)# interface fastethernet 0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# do show vlan
```



Step 5: Configure Trunk Mode on Inter-Switch Links

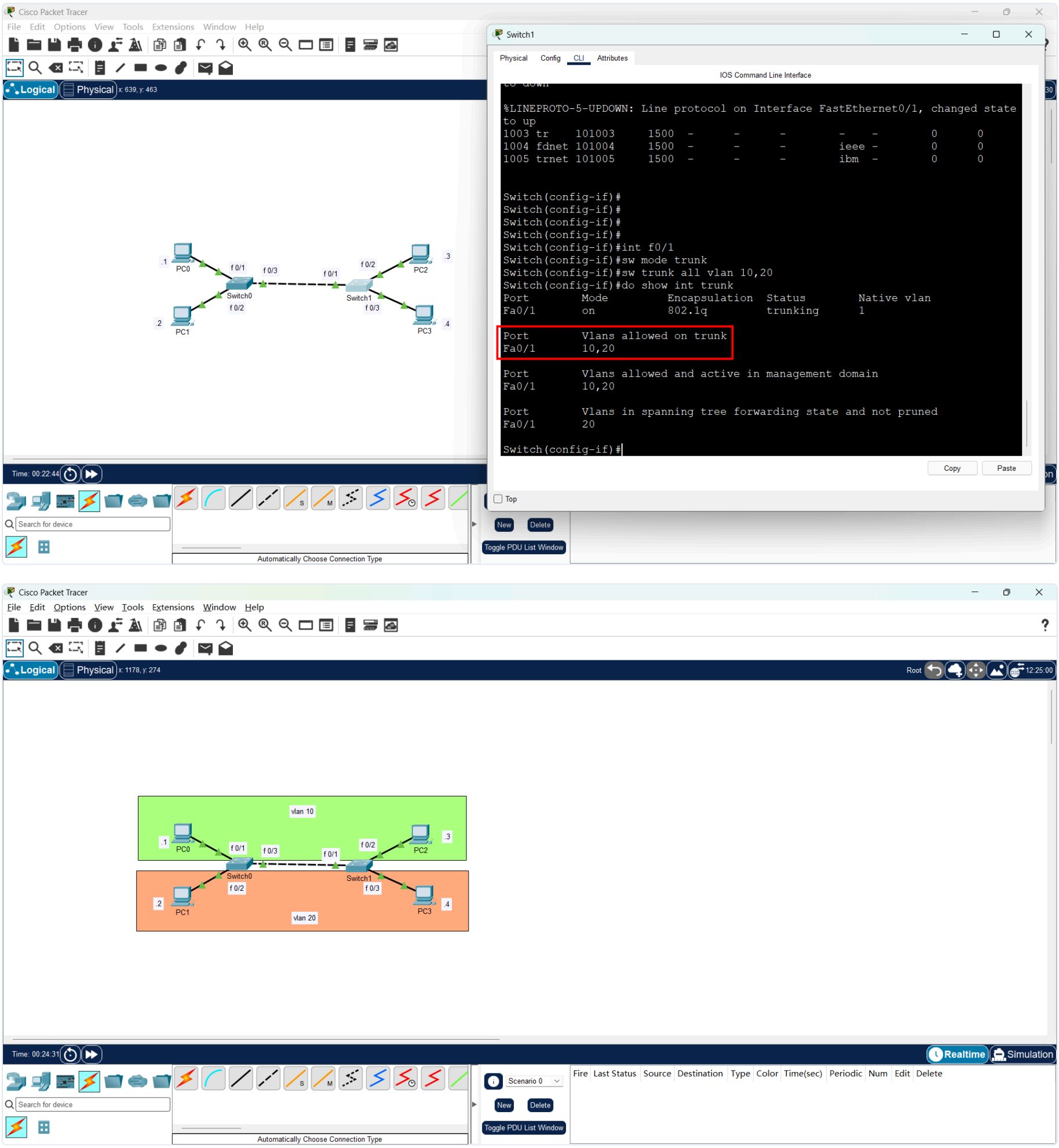
1. On **Switch0**, configure Fa0/3 as a Trunk link:

```
Switch(config)# interface fastethernet 0/3
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# do show interfaces trunk
```



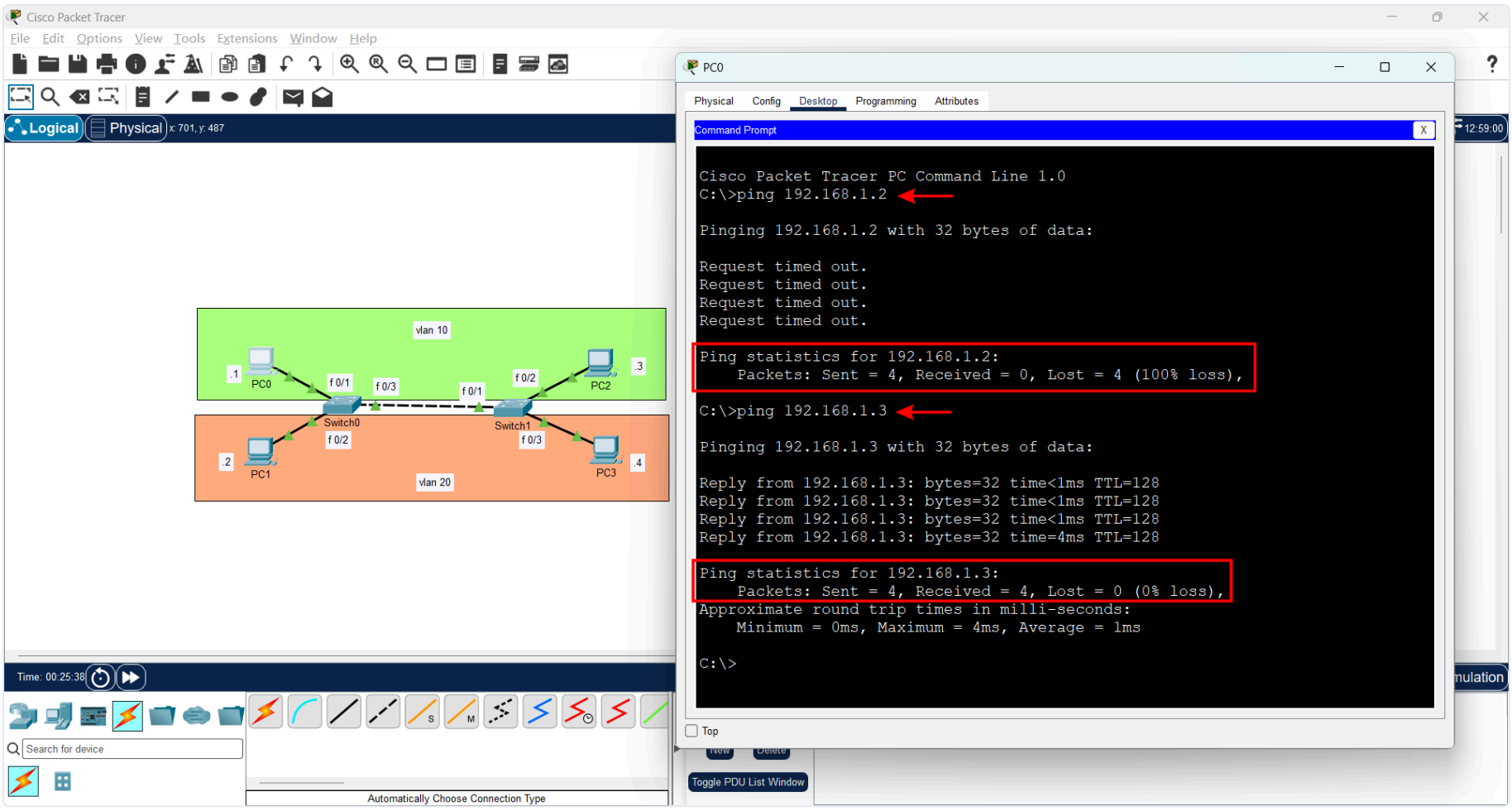
2. On **Switch1**, configure Fa0/1 as a Trunk link:

```
Switch(config)# interface fastethernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# do show interfaces trunk
```

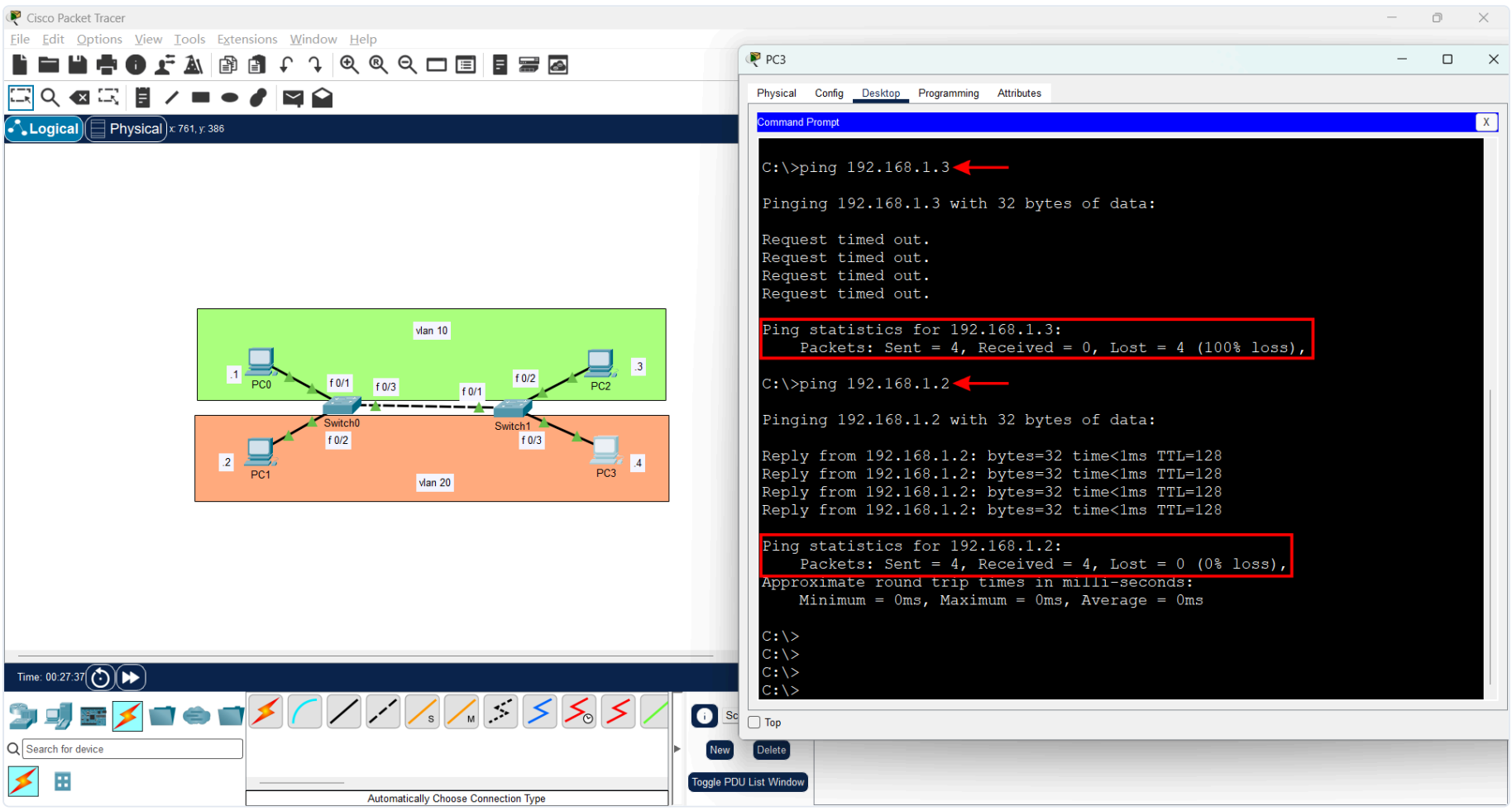


Step 6: Test Isolation and Connectivity

1. Open **PC0 (VLAN 10)** Command Prompt:
   - Ping PC2 (VLAN 10): `ping 192.168.1.3` → **Reply received (Success)**
   - Ping PC1 (VLAN 20): `ping 192.168.1.2` → **Request timed out (Blocked)**



2. Open **PC3 (VLAN 20)** Command Prompt:
   - Ping PC1 (VLAN 20): `ping 192.168.1.2` → **Reply received (Success)**
   - Ping PC2 (VLAN 10): `ping 192.168.1.3` → **Request timed out (Blocked)**



4. Lab 3 Questions & Analysis

Question 1: VLAN Broadcast Isolation

Although PC0 (192.168.1.1) and PC1 (192.168.1.2) are in the same IP subnet (192.168.1.0/24) and connected to the same switch, why do they fail to communicate? Explain the role of VLAN tags.

Question 2: Trunk Port Function

What is the primary difference between a switch port configured in Access mode versus Trunk mode? Why is a Trunk port necessary between Switch0 and Switch1?

Lab 4: Multi-Subnet Static Routing & Standard Access Control Lists (ACL)

1. Scenario & Objectives

In this lab, you will deploy a multi-router routed network with four distinct subnets and implement network security policies using Standard Access Control Lists (ACLs). You will configure static routing across two routers, verify baseline reachability, and apply ACL rules to filter traffic based on source IP addresses.

- Establish static routing across four distinct network segments via two intermediate routers.
- Verify complete baseline end-to-end reachability before deploying security filters.
- Configure Standard Named ACLs matching individual source hosts and entire subnets.
- Apply ACL filters to router outbound interfaces (`out`) and verify traffic permit/deny enforcement.

2. Subnet Architecture & Addressing Table

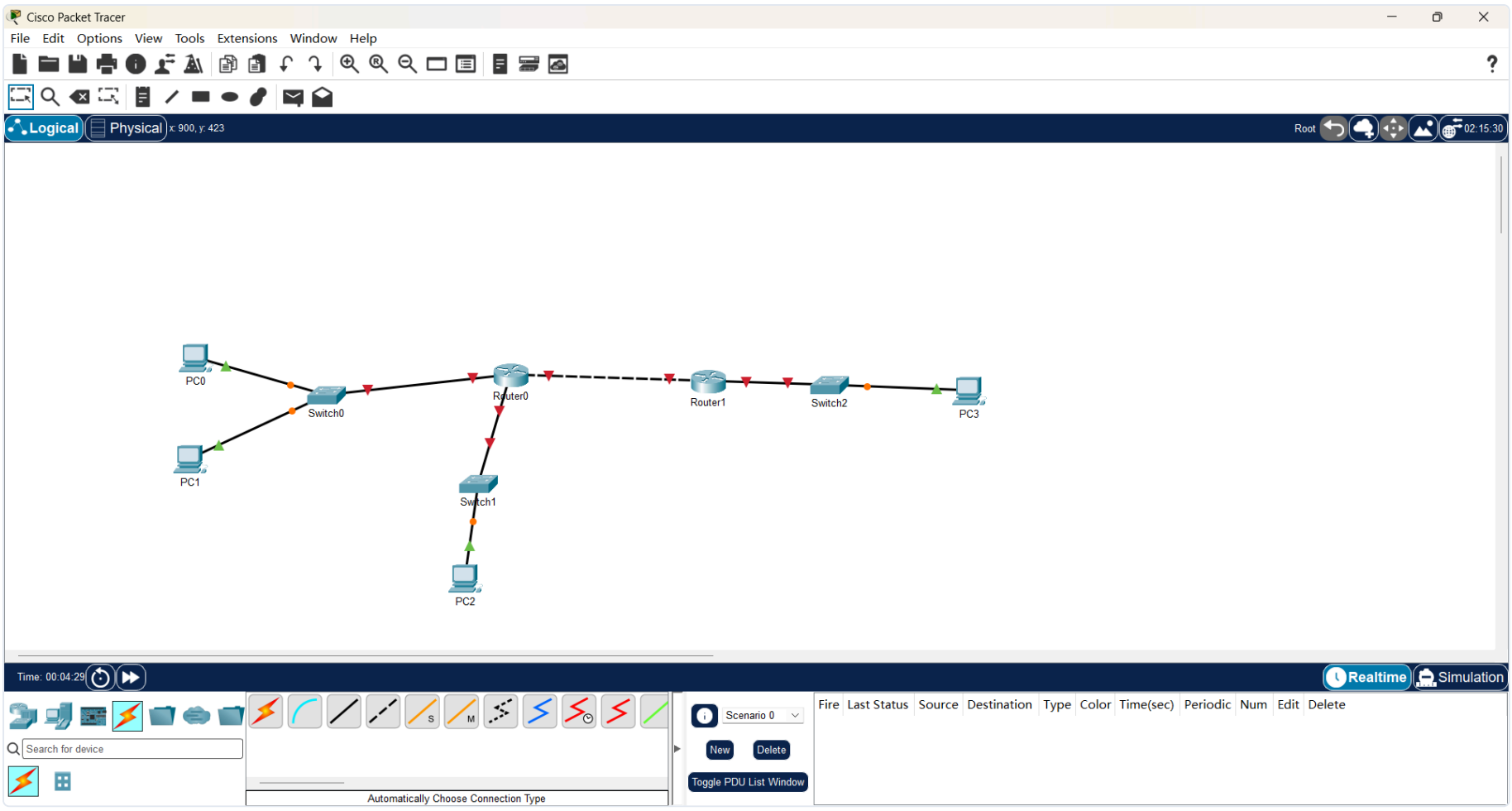
> **Four Subnet Partitions (/24 Prefix):**
> • **Orange Zone:** `192.168.1.0/24` | PC0 (.1), PC1 (.2), Router0 G0/0 (.254)
> • **Green Zone:** `192.168.2.0/24` | PC2 (.1), Router0 G0/1 (.254)
> • **Yellow Backbone:** `192.168.10.0/24` | Router0 G0/2 (.1) ↔ Router1 G0/0 (.2)
> • **Purple Zone:** `192.168.3.0/24` | PC3 (.1), Router1 G0/1 (.254)

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.1	255.255.255.0	**192.168.1.254**
PC1	FastEthernet0	192.168.1.2	255.255.255.0	**192.168.1.254**
PC2	FastEthernet0	192.168.2.1	255.255.255.0	**192.168.2.254**
PC3	FastEthernet0	192.168.3.1	255.255.255.0	**192.168.3.254**
Router0	Gig0/0	192.168.1.254	255.255.255.0	LAN Gateway for PC0/PC1
	Gig0/1	192.168.2.254	255.255.255.0	LAN Gateway for PC2
	Gig0/2	192.168.10.1	255.255.255.0	WAN Link toward Router1
Router1	Gig0/0	192.168.10.2	255.255.255.0	WAN Link toward Router0
	Gig0/1	192.168.3.254	255.255.255.0	LAN Gateway for PC3

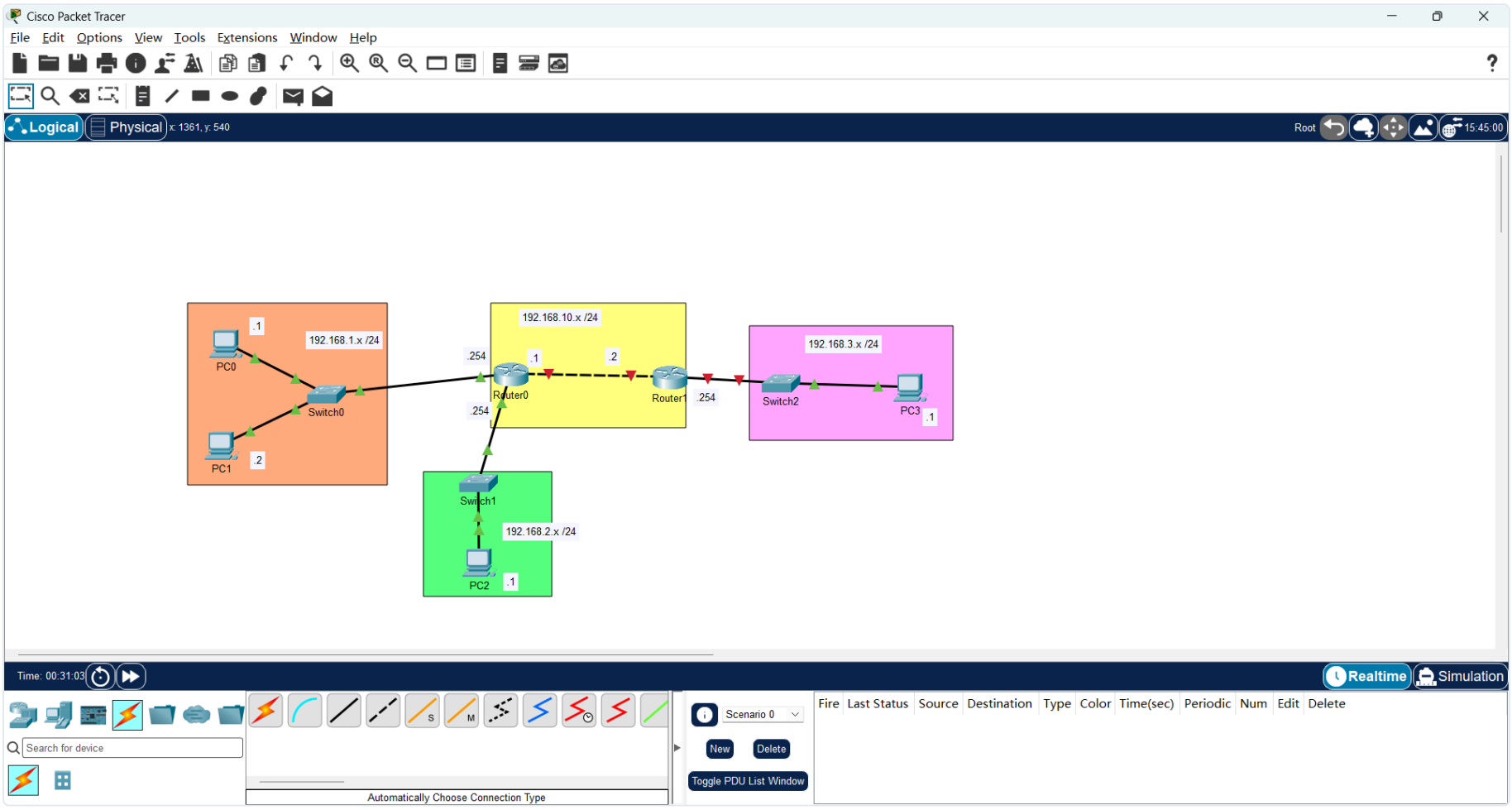
3. Step-by-Step Procedure

Step 1: Construct Topology and Configure Device IPs

1. Deploy 4 PCs, 3 switches, and 2 routers (Router0, Router1).
2. Use **Automatically Choose Connection Type** to connect devices.



3. Configure IP addresses and Default Gateways on all four PCs via the **Config** tab according to the addressing table.

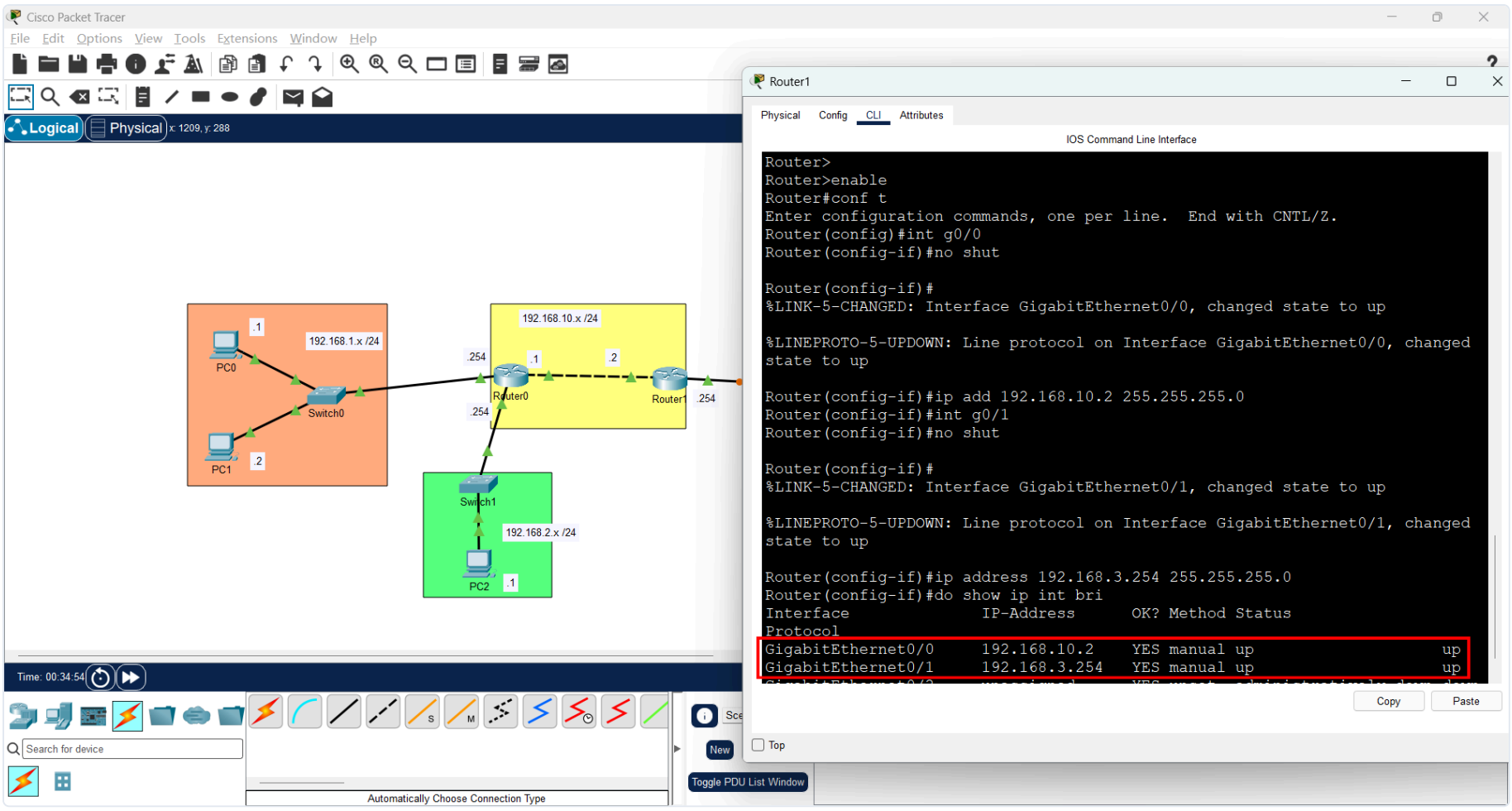


Step 2: Configure Router Interfaces and Static Routes

1. Open **Router1** CLI and configure interfaces:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0
Router(config-if)# ip address 192.168.10.2 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

Router(config)# interface gigabitethernet 0/1
Router(config-if)# ip address 192.168.3.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# do show ip int bri
```



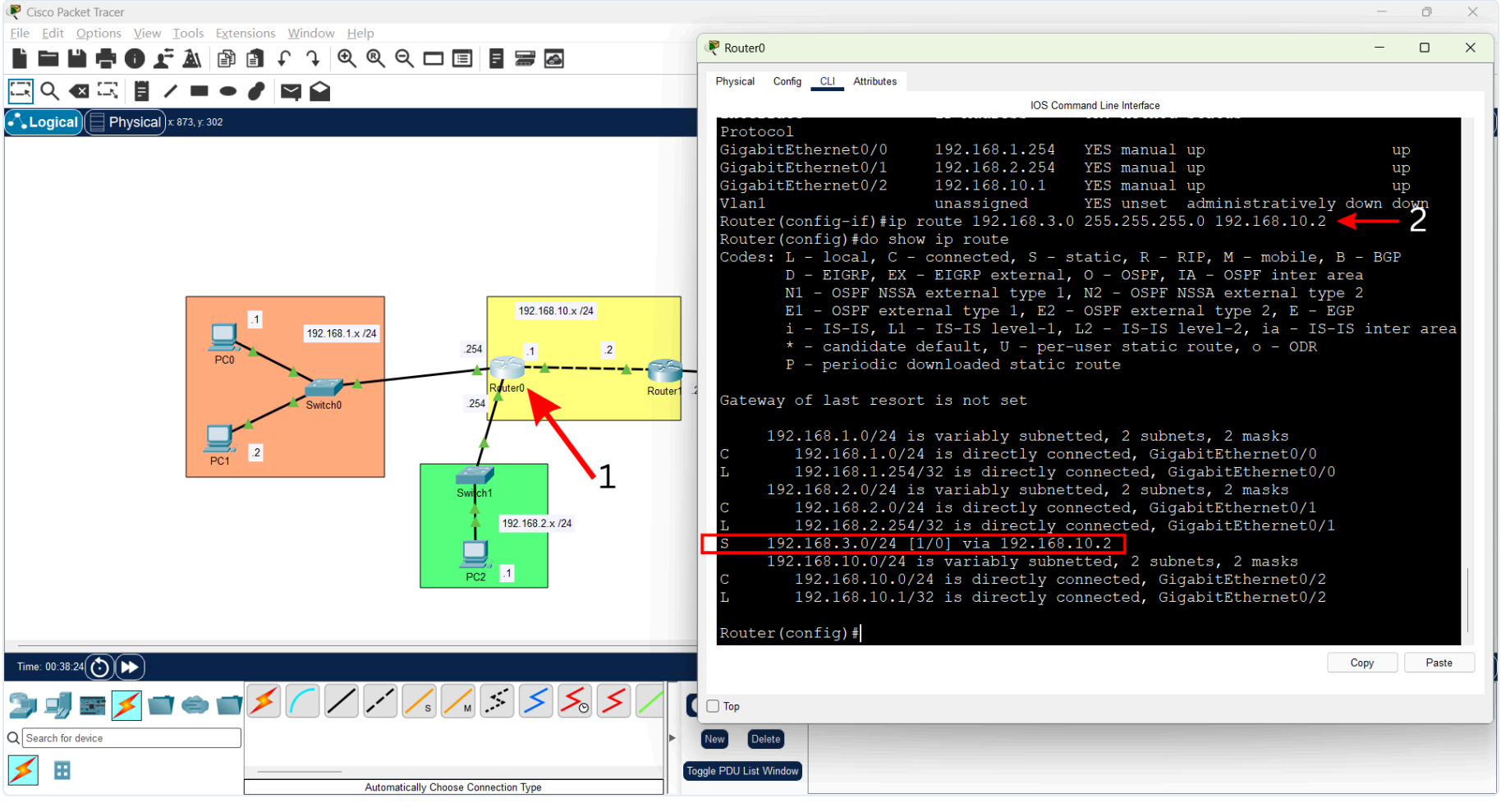
2. Open **Router0** CLI, configure interfaces, and add static route to Network 192.168.3.0/24:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitethernet 0/0
Router(config-if)# ip address 192.168.1.254 255.255.255.0
Router(config-if)# no shutdown

Router(config)# interface gigabitethernet 0/1
Router(config-if)# ip address 192.168.2.254 255.255.255.0
Router(config-if)# no shutdown

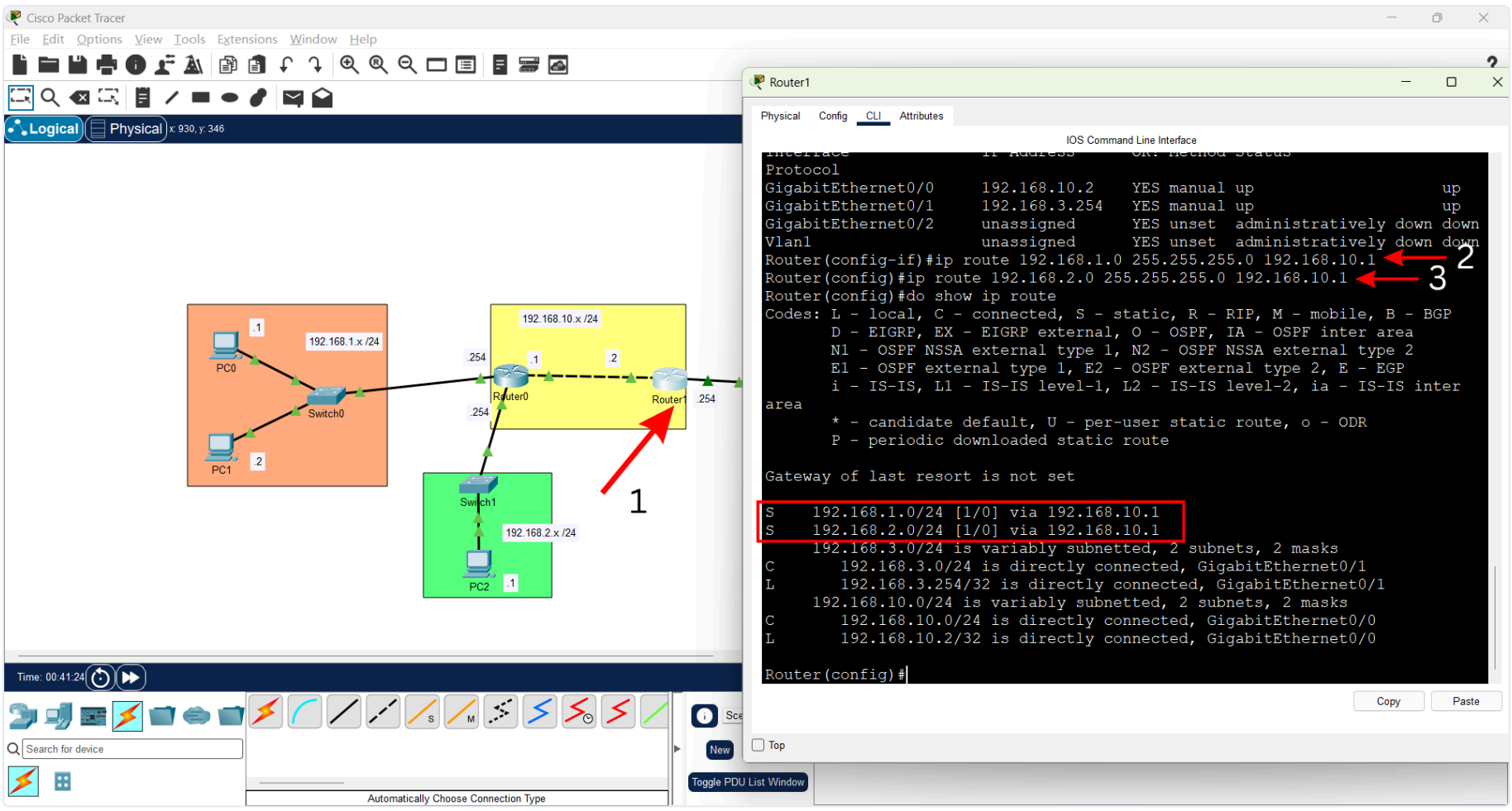
Router(config)# interface gigabitethernet 0/2
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Add static route to 192.168.3.0/24 via Router1
Router(config)# ip route 192.168.3.0 255.255.255.0 192.168.10.2
Router(config)# do show ip route
```



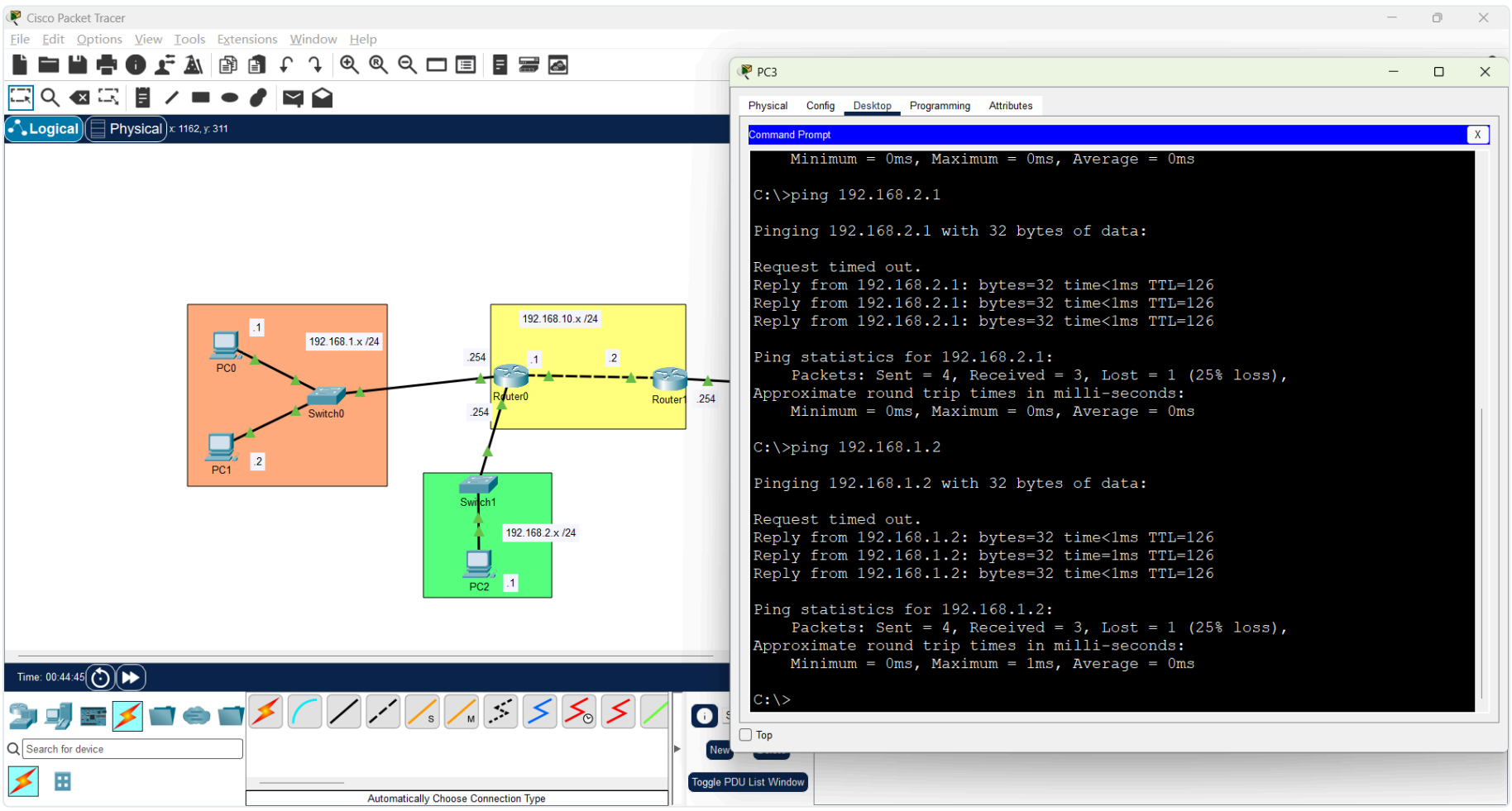
3. On **Router1**, add return static routes to Networks 192.168.1.0/24 and 192.168.2.0/24:

```
Router(config)# ip route 192.168.1.0 255.255.255.0 192.168.10.1
Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.10.1
Router(config)# do show ip route
```



Step 3: Baseline Connectivity Verification

1. Open **PC3** Command Prompt and verify initial baseline connectivity before applying ACLs:
   - `ping 192.168.2.1` (PC2) → **Reply received (Success)**
   - `ping 192.168.1.2` (PC1) → **Reply received (Success)**

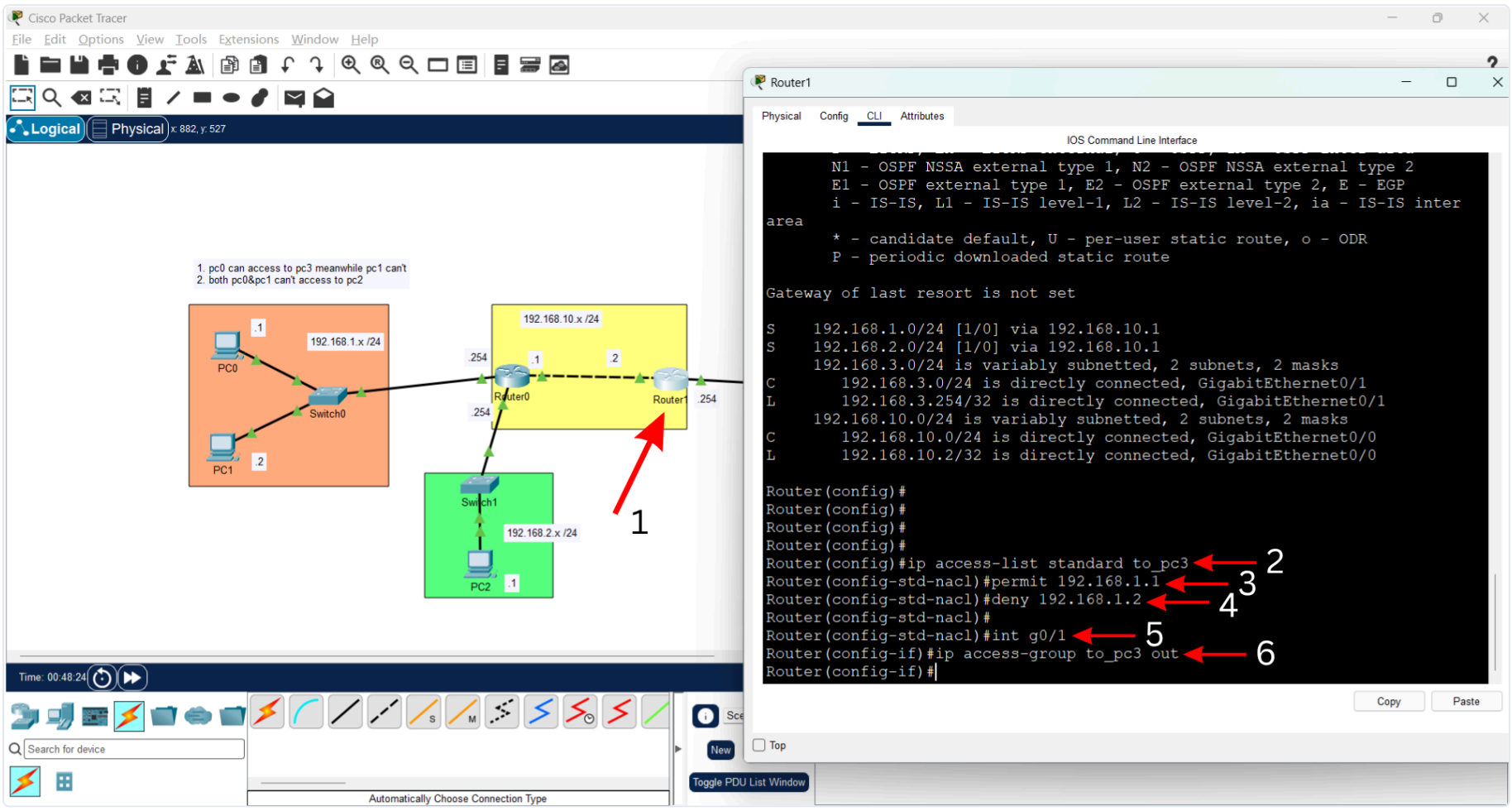


Step 4: Configure and Apply Standard ACL Rules

> **Security Policy Requirements:**
> - **Policy 1:** PC0 can access PC3, while PC1 cannot.
> - **Policy 2:** Neither PC0 nor PC1 (entire 192.168.1.0/24 subnet) can access PC2.

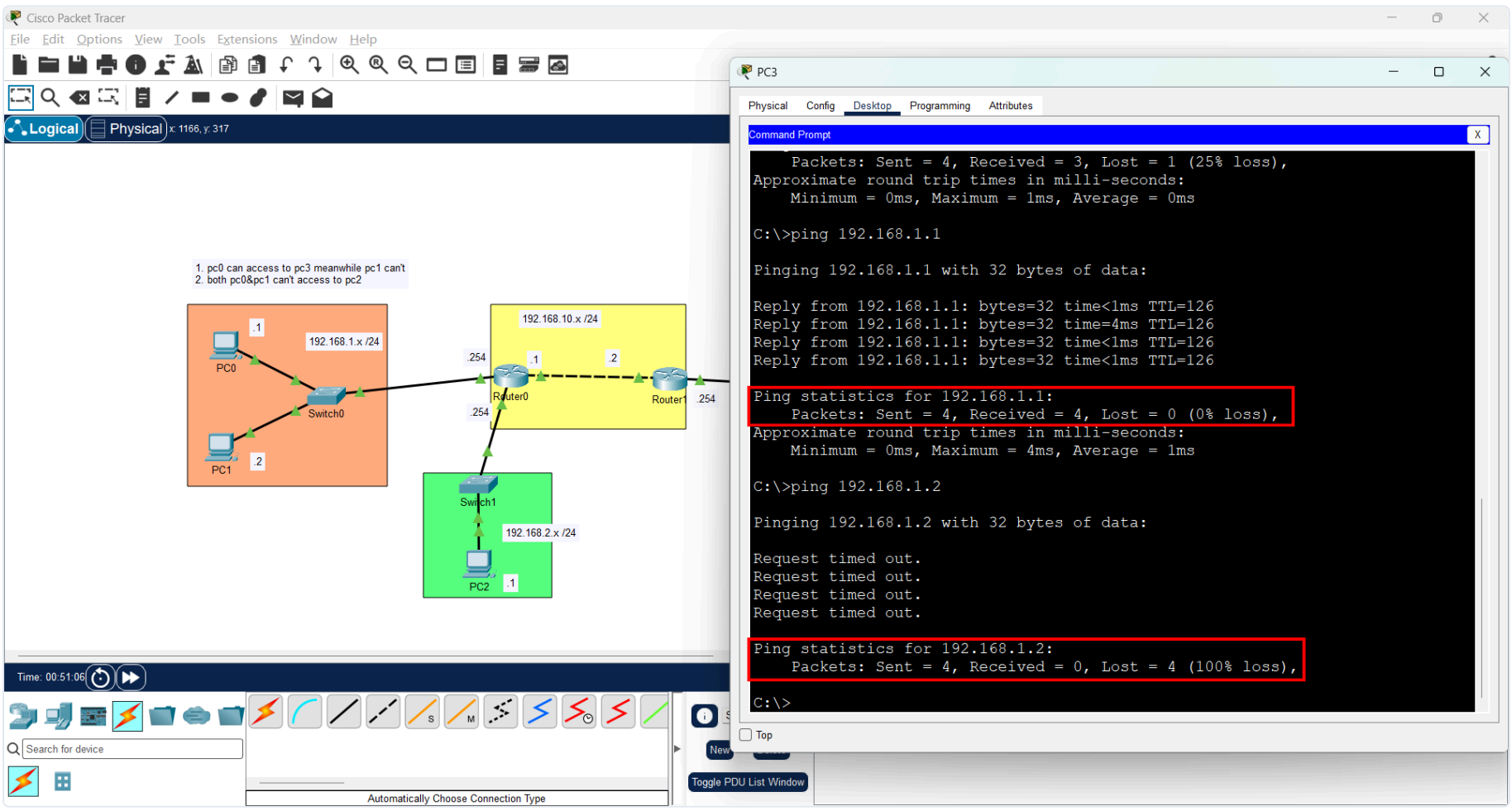
1. On **Router1**, create and apply ACL `to_pc3` :

```
Router1(config)# ip access-list standard to_pc3
Router1(config-std-nacl)# permit 192.168.1.1
Router1(config-std-nacl)# deny 192.168.1.2
Router1(config-std-nacl)# exit
Router1(config)# interface gigabitethernet 0/1
Router1(config-if)# ip access-group to_pc3 out
Router1(config-if)# exit
```



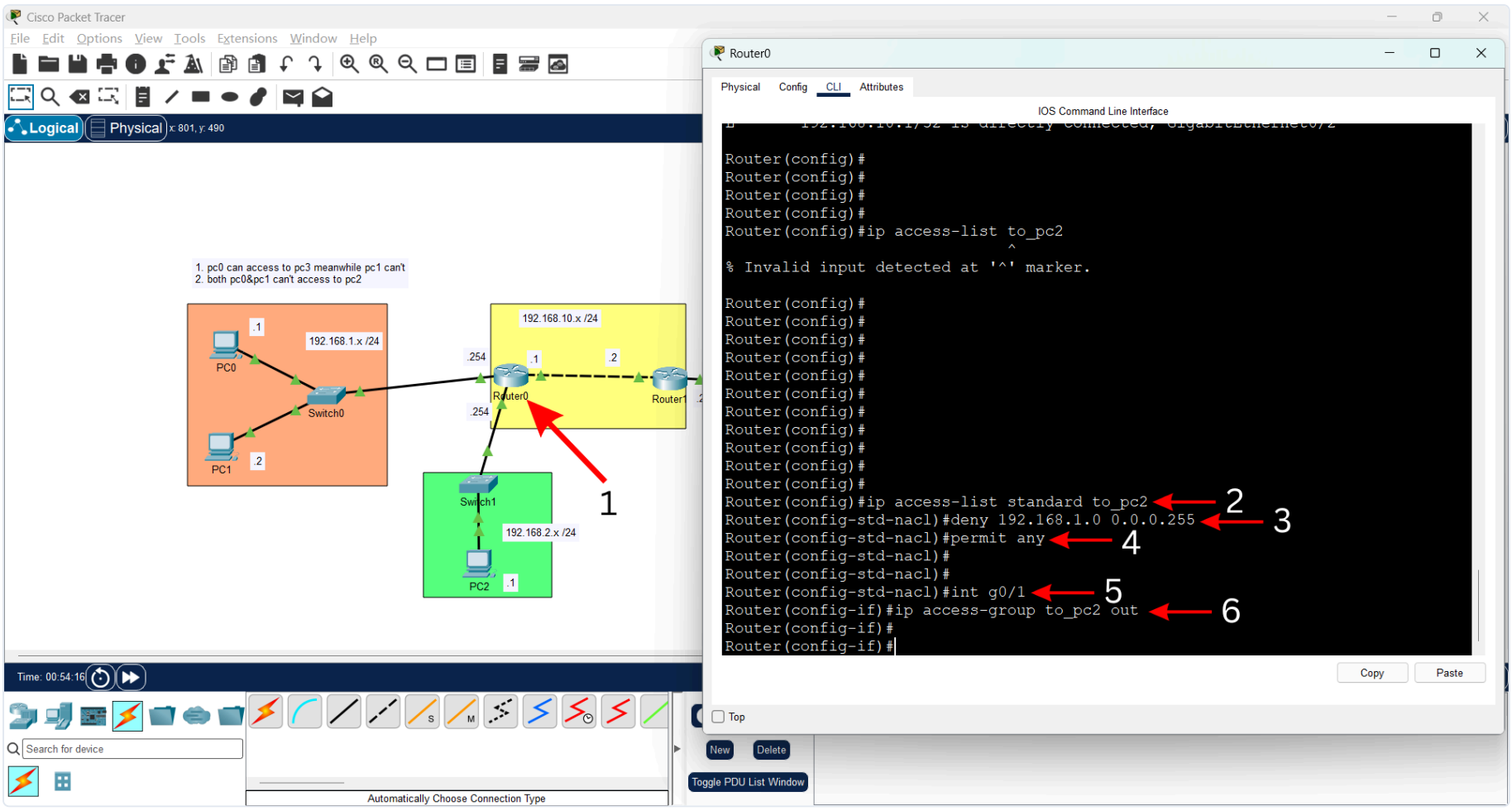
2. On **Router0**, create and apply ACL `to_pc2` (using wildcard mask `0.0.0.255`):

```
Router0(config)# ip access-list standard to_pc2
Router0(config-std-nacl)# deny 192.168.1.0 0.0.0.255
Router0(config-std-nacl)# permit any
Router0(config-std-nacl)# exit
Router0(config)# interface gigabitethernet 0/1
Router0(config-if)# ip access-group to_pc2 out
Router0(config-if)# exit
```

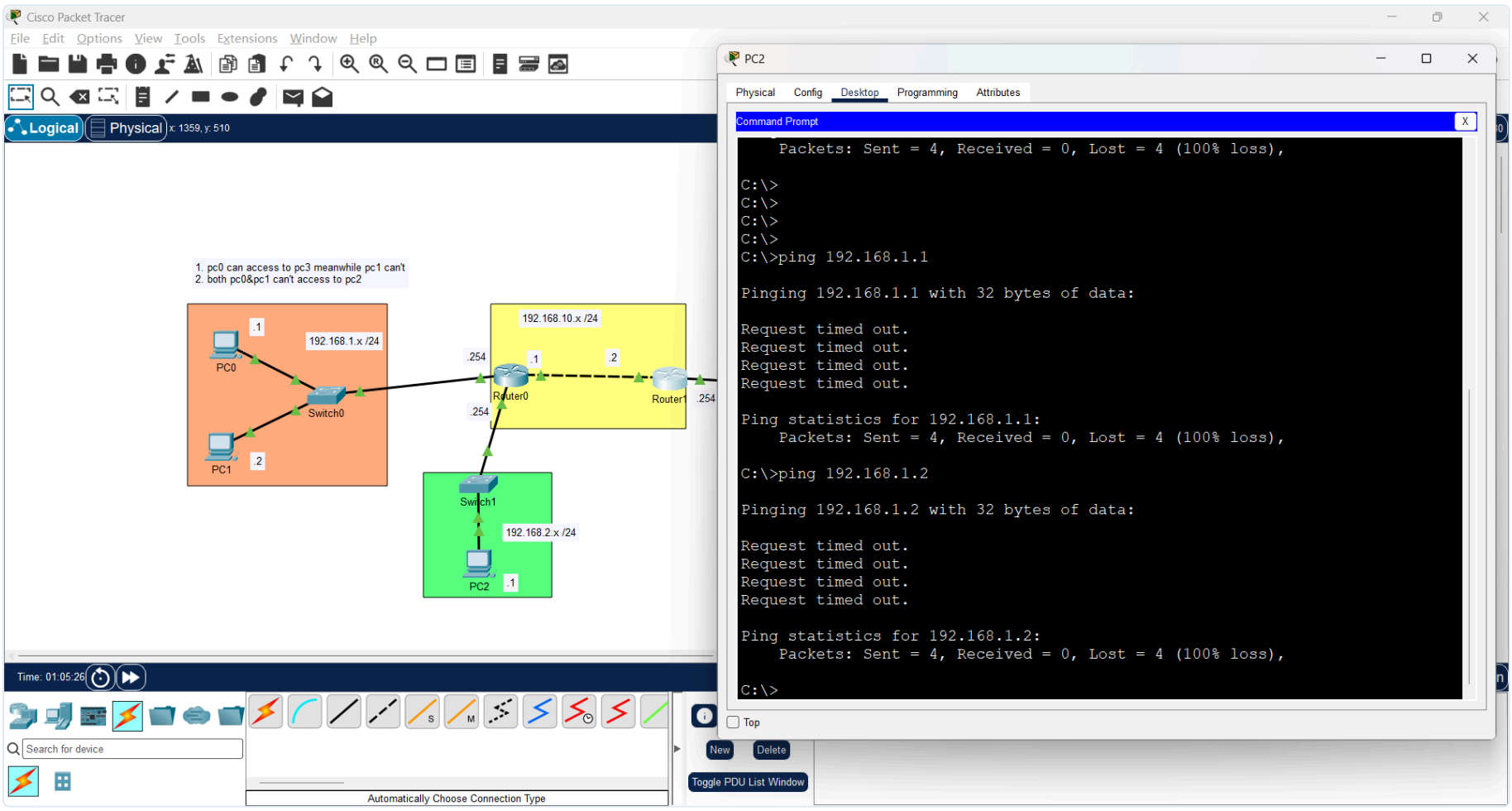


Step 5: Verify ACL Security Enforcement

1. **Verify Policy 1 on PC3:**
   - `ping 192.168.1.1` (PC0) → **Reply received (Success)**
   - `ping 192.168.1.2` (PC1) → **Request timed out (Blocked)**



2. **Verify Policy 2 on PC2:**
   - `ping 192.168.1.1` (PC0) → **Request timed out (Blocked)**
   - `ping 192.168.1.2` (PC1) → **Request timed out (Blocked)**



4. Lab 4 Questions & Analysis

Question 1: Standard ACL Placement Rule

Why are standard ACLs typically placed as close to the destination as possible (e.g., outbound on the target LAN interface) rather than near the source?

Question 2: Implicit Deny Behavior

At the end of every Cisco Access Control List, there is an invisible implicit rule. What is this default rule, and why was `permit any` required in ACL `to_pc2` ?

5. Comprehensive Troubleshooting Guide

Symptom	Likely Root Cause	Recommended Action
Router port LED stays Red	Interface is administratively shut down.	Enter interface configuration mode and execute `no shutdown` .
Destination host unreachable	Host Default Gateway is unconfigured or router has no route to the destination.	Check host IP configuration for correct gateway IP; execute `show ip route` on the router.
Request timed out across routers	Missing return static route on the remote router.	Verify that both forward and return routes exist along the entire communication path.
VLAN hosts cannot ping across switches	Inter-switch connection is in Access mode or required VLANs are not allowed on the Trunk.	Run `show interfaces trunk` and verify the port is in Trunk mode with the VLANs allowed.
Unintended traffic blocked by ACL	Implicit deny statement dropped traffic that was not explicitly permitted.	Add a `permit any` statement at the end of the ACL if only specific subnets should be denied.